Novel Rule-Based Annotation Models for Comparative Pangenomics

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Abstract

The evolution of comparative genomics prompted a shift towards the study of prokaryotic diversity at the level of pangenomes. This led to the development of diverse pangenome construction and analysis methodologies over the years. PPanGGOLiN stands out with an innovative methodology integrating both gene families co-occurence patterns and a gene-level graph architecture based on genomic contiguity to partition the pangenome. This allows for efficient exploitation of genomic contexts paving the way for functional annotations and comparative analyses at unprecedented scales, with particular relevance to the study of prokaryotic diversity and evolutionary dynamics. To harness this potential, PANORAMA was developed to annotate macromolecular systems and conduct pangenomic comparative analyses on the pangenome graph outputs of PPanGGOLiN. However, PANORAMA remains currently restricted to the prediction of bacterial defense systems.

PULs are molecular systems responsible for the degradation of complex carbohydrates, notably known to be characteristic of certain Bacteroidetes that make part of the human gut microbiome. PULs have been the subject of many studies over the years, with multiple PUL databases and prediction methods developed in recent years. Nonetheless, current PUL annotation methods remain restricted to the genome-level, which inherently limits their scope and effectiveness. The integration of new rule-based models for PUL prediction into PANORAMA framework promises to offer several advantages, including improved sensitivity and specificity of PUL detection across diverse species and enabling deeper biological insights through comparative analysis at the pangenome level. All scripts used for data analysis can be accessed in the GitHub repository: https://github.com/yazid-hoblos/PULkit.

Keywords: pangenome, comparative pangenomics, rule-based annotation, Bacteroidetes, CAZymes, polysaccharide utilization loci (PUL)

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Glossary

HGT Horizontal Gene Transfer

BMM Bernoulli Mixture Model

HMM Hidden Markov Model

HMRF Hidden Markov Random Field

GI Genomic Island

RGP Regions of Genomic Plasticity

CAZyme Carbohydrate-Active enZYme

CGC CAZyme Gene Cluster

PUL Polysaccharide Utilization Locus

Sus Starch utilization system

MSA Multiple Sequence Alignment

PPanGGOLiN Partitioned PanGenome Graph Of Linked Neighbors

1 Introduction

Advancements in sequencing technologies led to a substantial increase in the number of available prokaryotic genomes, with comparative genomics analyses revealing extensive intra-species genomic diversity [1]. In 2005, Tettelin et al. [2] proposed the pangenome concept to account for the significant variations observed between the genomes of different strains of the same prokaryotic species. These differences are the outcome of the special genome dynamics that characterize prokaryotes, particularly gene gain from other species through horizontal gene transfer (HGT) and differential gene loss [3]. The pangenome represents the gene content of a given set of genomes of the same species; it is composed of *core* genes shared across all genomes and *accessory* genes present only in some of the genomes [4]. However, strict definitions of the core genome lead to constant reduction of its size as more genomes are considered, mostly due to technical artifacts (i.e. incomplete genome sequences). Thus, it was proposed to alternatively divide the pangenome into a *persistent* genome, consisting of the gene families present in almost all genomes, a *shell* genome, consisting of families of intermediate frequencies in the species, and a *cloud* genome for families present at low frequencies in the species [5].

Pangenome construction and analysis methodologies have evolved significantly over the years. Classical approaches rely on predefined thresholds to partition the pangenome into its core and accessory components [6, 7, 8]. These include a sequence identity threshold to group genes into families and a presence cut-off specifying the proportion of genomes in which a family has to exist in order to be considered as part of the core genome. Other approaches rely on statistical methods to define the different components of the pangenome. For instance, van Tonder et al. [9] uses a Bayesian model, whereas Snipen et al. [10] defines a Binomial mixture model for pangenome partitioning. Furthermore, to account for the significant variations in individual reference genomes, pangenome graphs were suggested to model a sequence-level consensus representation of many genomes by defining an alternative path for each known nucleotide variation [11, 12]. More recently, PPanGGOLiN (Partitioned PanGenome Graph Of Linked Neighbors) was proposed with an innovative methodology making use of different aspects of previous approaches, by integrating both a statistical Bernoulli Mixture Model (BMM) and a gene-level pangenome graph architecture for pangenome partitioning [13]. PPanGGOLiN uses a graph structure based on genetic contiguity, where nodes represent gene families and edges indicate adjacency of the two connected families on a contig. The edges are weighted by the number of genomes in which the relation of contiguity exists. Unlike sequence-level pangenome graphs, the gene-level graph of PPanGGOLiN provides better biological interpretability and the possibility to exploit genomic contexts, such as identifying conserved operons and understanding functional associations in genomic neighborhoods. PPanGGOLiN combines its BMM model, defined over gene families presence-absence matrix, with a Hidden Markov Random Field (HMRF), defined over the graph structure, to partition the pangenome graph into *persistent*, *cloud*, and *shell* components. Notably, The pangenomes of all prokaryotic genomes in GenBank—clustered according to GTDB taxonomy—were constructed with PPanGGOLiN to create PanGBank database as a comprehensive resource for prokaryotic pangenomes [14]. Moreover, the PPanGGOLiN software suite has been extended with two methods, PanRGP [15] and PanModule [16], which leverage PPanGGOLiN graph architecture to detect Genomic Islands (GI) and their corresponding conserved modules, thus allowing the study of GIs and their diversity on an unprecedented scale of hundreds or thousands of genomes.

Prokaryotic cells are maintained by a set of macromolecular systems, which include complex nanomachines, such as ribosomes, as well as the diverse molecular pathways underlying cellular metabolism [17, 18]. Macromolecular systems encompass diverse functional categories, ranging from metabolic pathways to defense mechanisms and secretion systems. The genes that make up these systems are typically organized in close proximity in the form of operon structures [19]. Polysaccharide utilization loci (PUL) are one such important macromolecular system responsible for the degradation of complex carbohydrates, such as starch and cellulose [20]. They are typical of the *Bacteroidetes* phylum, notably known to be central to the human gut microbiota. However, they have been increasingly identified in other bacterial phyla and some Archaea, often exhibiting distinct organizational patterns that deviate from the canonical features of *Bacteroidetes* PULs [21].

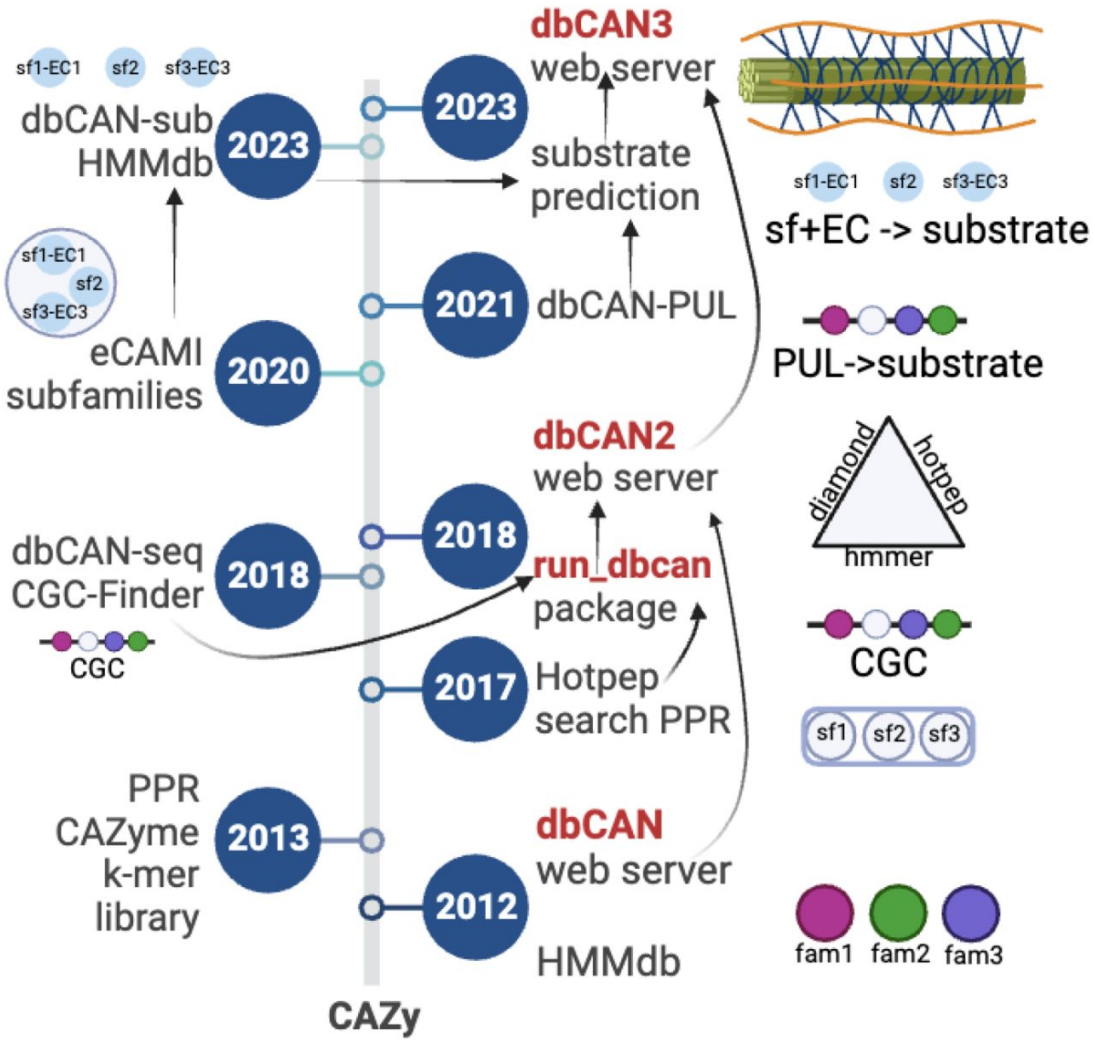


Figure 1: Timeline of dbCAN evolution over the years [22].

The classification and study of PULs and their constituent carbohydrate-active enzymes (CAZymes) have evolved significantly over the years. In 2009, the Carbohydrate-Active EnZymes database (CAZy) was first introduced as a comprehensive resource for all enzymes that build and breakdown complex carbohydrates [23]. This allowed the development of dbCAN in 2012, as a tool for the prediction and annotation of CAZymes based on defined signature domains for every CAZyme family [24]. Since then, dbCAN has undergone substantial extensions over the years, most notably the addition of a feature for CAZyme gene clusters (CGCs) prediction as putative PULs in dbCAN2 release in 2018 [25]. Thereafter, a specialized database for empirically verified CGCs and their substrates was introduced under the name dbCAN-PUL in 2020 [21]. The latest version, dbCAN3, was introduced in 2023 with focus on substrate predictions [26]. This timeline of dbCAN evolution over the years is summarized in Fig. 1. In parallel, the team responsible for the development of CAZy database separately introduced a model for automatic prediction of PULs in Bacteroidetes in 2015 [27], and used it to construct CAZy-PULDB as a resource for PULs derived from predictions in Bacteroidetes species isolated from the human gut as well as reported experimental findings [28].

Annotation of bacterial macromolecular systems has been the subject of growing interest over recent years. MacSyFinder is one of the earliest tools to develop a framework for rule-based systems prediction [18]. Since then, several specialized rule-based tools were developed for the prediction of specific functional systems, such as PADLOC [29] and DefenseFinder [30] for bacterial defense systems, and PULpy [31] and dbCAN2 [25] for PULs. While all these approaches remain at the scale of single genomes, PANORAMA was recently introduced to perform rule-based predictions and comparative analysis directly on the pangenome graph architecture of PPanGGOLiN (Arnoux et al., in preparation). With these innovations, PANORAMA introduces a formalized framework for comparative pangenomics, paving the way for unprecedented scales of analysis and potential new insights into bacterial evolutionary dynamics.

In this project, we aim to integrate novel rule-based models for PUL prediction into the framework of the *PanSystem* workflow of PANORAMA. To this end, PUL models will be formulated based on previously reported empirical findings and prediction models, as captured in the CAZy and dbCAN resources. This will be followed by comparative analysis with existing benchmark methods, such as PULpy [31] and dbCAN2 [25], to illustrate the advantages of direct annotation on the pangenome-level relative to working at the level of single genomes. Subsequently, the *PanCompare* workflow of PANORAMA will be used to analyze the identified PULs across different species to draw relevant biological insights.

The specific questions to be addressed are the following:

1. Can PULs be reliably annotated at the pangenome level using rule-based models?
2. Does annotation on pangenome graphs improve the sensitivity and specificity of PUL detection compared to traditional single-genome methods?
3. How well does the newly introduced PUL model perform across diverse species and pangenome sizes?
4. What biological insights could be revealed from comparative analysis of identified PULs across different species and taxonomic groups?

2 Materials & Methods

2.1 PANORAMA Workflow & Modules

PANORAMA consists of two main workflows: *PanSystem* for rule-based annotation of biological systems on pangenome graphs, and *PanCompare* for comparison of biological contexts across pangenome graphs. Both workflows use the pangenome H5 files generated by PPanGGOLiN. The different steps of the *PanSystem* workflow are highlighted in Fig. 2. It starts by annotating representative sequences of all gene families using predefined Hidden Markov Model (HMM) profiles. Then, the model rules are checked in different genomic contexts, with a transitivity parameter defining the maximum allowed genomic distance between two gene families within the system. Finally, the predicted macromolecular systems are projected back onto the pangenome graph.

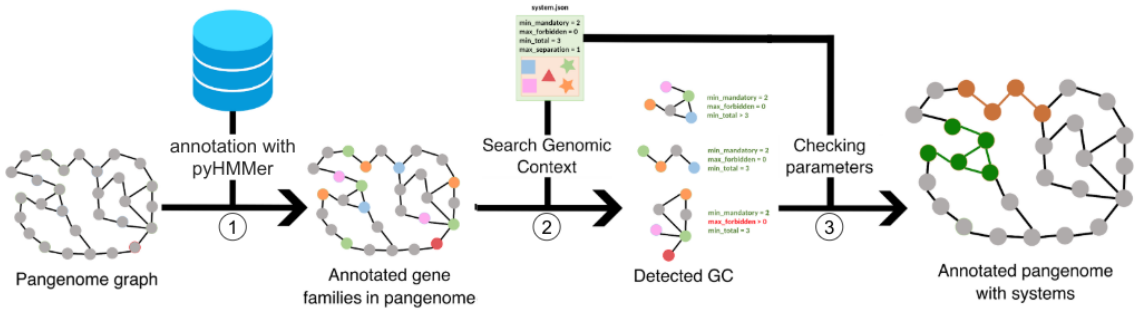


Figure 2: The sequence of steps performed by the *PanSystem* workflow of PANORAMA for macromolecular systems prediction. pyHMMer is first used for functional annotation of gene families, then relevant genomic contexts are extracted and the rules of the predefined models are examined in them. Detected systems are added to the pangenome annotation and projected back onto individual genomes (Source: Arnoux et al., in preparation).

Gene families are annotated based on HMM profiles, as typically found in Pfam entries of protein families. The main steps of profile HMM construction are depicted in Fig. 3. HMMs are intended to capture the statistical characteristics of conserved protein domains so as to allow for the identification of new proteins that match these patterns. As highlighted in Fig. 3, the process begins with a multiple sequence alignment (MSA) of homologous proteins, which is then used to guide the construction of a probabilistic graph encoding the amino acid frequencies at each position, while also accounting for insertions and deletions.

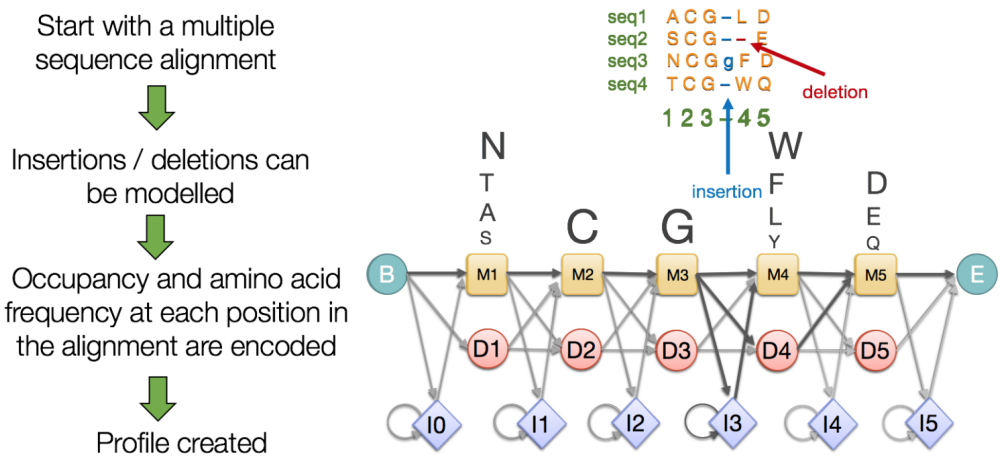


Figure 3: An example illustrating the construction of a profile Hidden Markov model (HMM) for protein annotation. A multiple sequence alignment (MSA) of homologous proteins is used to guide the construction of a probabilistic graph that encodes the frequencies of the different amino acids, as well as the likelihood of insertions and deletions, at each position (Source: N. Mohd et al., 2019).

Rule-based annotation is guided by a set of predefined instructions based on reported empirical findings. Figure 4 highlights an example of how PANORAMA applies rule-based predictions on its extracted genomic contexts. As shown, this involves two main hierarchical units: isofunctional gene families and functional units (FUs), defined as groups of families that perform a system functionality together. Moreover, each defined model includes a transitivity parameter, constraining the genomic organization of the system. Families and FUs are categorized as mandatory, accessory, forbidden, or neutral. PANORAMA extracts genomic contexts and examines the rules of the different predefined models to guide its predictions. In the highlighted example, we can see that this leads to the prediction of models A and B, while models C and D are not predicted due to the detection of a family forbidden in them.

In the context of this project, this workflow of PANORAMA was leveraged to conduct rule-based PUL predictions on pangenome graphs, successfully addressing our first scientific question Q1. All pangenomes used for PUL detection were accessed from the PanGBank database [14] of precomputed PPanGGOLiN pangenomes. For each pangenome, all corresponding genomes used to construct it were accessed from NCBI to run dbCAN predictions on them. In a few cases, pangenomes were recomputed from these genomes to ensure consistent annotations, since the annotation of some genes is constantly being updated on NCBI. Alternatively, the protein FASTA and annotation GFF files could be extracted from PPanGGOLiN pangenomes to run dbCAN on them. A list of all Prevotella species and their corresponding strains used in our analysis could be found in Table 1 in the appendix.

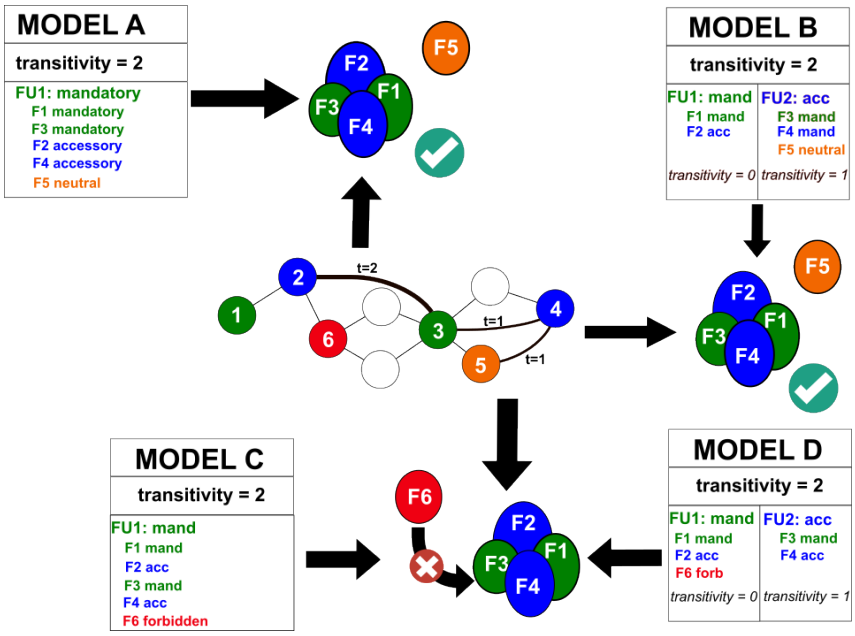


Figure 4: A simplified illustration of the rule-based prediction approach of PANORAMA. The rules of four models are examined in the highlighted genomic context, extracted from a PPanGGOLiN pangenome graph. The rules include a transitivity parameter, min_mandatory and min_total parameters for the minimum number of mandatory elements and accessory elements in a system, and a series of families and functional units (FUs) categorized as mandatory, accessory, forbidden, or neutral (Source: Arnoux et al., in preparation).

2.2 Model Formulation

Two primary PUL prediction models could be identified in the literature. The first, introduced by Terrapon et al. [27] in 2015, defined PULs as operon-like structures centered around a SusC-SusD adjacent pair. Thus, this initial model relied on a few *Pfam* domains to annotate all SusC/D-like proteins, where SusC refers to TonB-dependent transporters and SusD refers to surface glycan-binding lipoproteins that help deliver complex polysaccharides to SusC. Subsequently, the approach identifies tandem pairs of SusC-D and uses the operon databases, DOOR [32] and ProOpDB [33], to guide the extension of predicted PULs to include adjacent genes that are likely to be working in an operon together. This approach was used to create the initial version of CAZy-PULDB. It primarily concerned Bacteroidetes species as it was defined and validated based on them. Its accuracy was further established through their extensive application to more than 5000 Bacteroidetes genomes in PULpy [31]. Moreover, in 2018, Cazy-PULDB was updated to account for additional PUL types. The rules were updated to apply exactly the same previous algorithm, but instead of initiating the prediction around susCD genes, it starts from a core of at least three adjacent CAZyme genes, not necessarily on the same strand, separated by a maximum of one non-signature gene. They also expanded their annotation of PUL components to functionally relevant proteins in their vicinity, such as sulfatases and proteases.

Inspired by this initial approach, a second PUL prediction model was introduced as part of dbCAN2 in 2018 [25]. In an attempt to further generalize PUL systems definition, the new CGC-Finder approach initially defined CAZyme Gene Clusters (CGCs) as clusters of at least one CAZyme, one transporter component (TC), and one transcription factor (TF), separated by a maximum of two non-signature genes. This attempt was primarily based on their previous

success in developing a well-established CAZymes annotation approach, integrating HMM defined profiles and sequence similarity searches against CAZy. Thus, they extended their CAZyme annotations with TC and TF annotations based on diamond blasts against TCDB and TF databases, such as CollectTF and DBTBS, and complemented this with the use of the genomic coordinates of gff files to return PUL-like predictions. Afterwards, the approach added Signal Transduction Proteins (STPs) as a fourth class of signature genes which were annotated using a set of HMM profiles appended to dbCAN HMM database. It also shifted to using TF HMM profiles instead of diamond for the annotation of TFs, leaving only TC annotations for diamond.

Given its more generic nature and the comprehensive set of CAZyme HMM profiles provided by dbCAN to support it, the CGC-Finder model was initially adopted to guide PANORAMA PUL predictions. Accordingly, it was formulated in the JSON rules format used by PANORAMA and adopted from MacSyFinder [18], which first formalized rule-based systems prediction. This simply entailed defining a minimum number of 3 mandatory families (CAZyme, TF, TC) as well as one accessory family (STP). Each of these classes is associated to all HMM profiles, or diamond results in the case of TCs, which are all interpreted equivalently by the model. Alternatively, if the gene families are directly annotated as these 4 classes, the model could be simplified to drop these long association lists. Finally, a transitivity parameter (t) is defined to dictate how far each annotated signature gene family can reach during systems detection. Thus, at t=0, each family will be only connected to the two directly next to it; at t=1, it can reach for two more, and so on. This also involves a window size (w) parameter that is defined by default as t+1, but could be redefined in the model to be higher. This allows each signature family to consider the w genes next to it, then for each t+1 edges will be established with its neighboring genes. Therefore, while not directly corresponding to it, these two parameters were used to emulate dbCAN distance parameter, which is defined as the maximum number of non-signature genes allowed between two signature genes. But their influence goes beyond this simple linear effect since they are applied to the complex structure of a pangenome-level context graph.

2.3 Source Code Issues

A significant challenge in using PANORAMA was the lack of documentation for its functionalities and output formats, as well as the complexity of the source code organization since it was still under development. Even after managing to successfully obtain some initial system predictions, they were inconsistent over different runs, very challenging to interpret, and showed significant discrepancy relative to dbCAN predictions, both at the level of the number of predicted systems and their content. Many genes grouped within one PUL by PANORAMA were scattered across multiple CGCs in dbCAN predictions, in addition to major differences at the level of all proteins predicted to be part of PULs between the two methods. Moreover, there were multiple paradoxical observations when considering PANORAMA predictions on their own. For instance, across all genomes, many families were found to correspond to so many different predicted PULs, certain families found in the genome projections of certain systems were not found in the output file of the systems themselves, and the systems extended for very long ranges even at low transitivity values.

Later on, I also discovered an error in dbCAN CGC-Finder code that was undermining their whole approach. The approach was implemented to create a reduced GFF file from the original one to include annotations of the signature gene categories (CAZyme, TF, TC, and STP), while

leaving all other annotation empty, so that once more than a given number of non-signature genes is encountered between signature genes, the extension of one CGC stops and another one begins. Surprisingly, a bug in the code was skipping all non-signature genes leading to a GFF file with only the signature genes. Accordingly, the only reason not all signature genes were classified within one CGC was because dbCAN allows the user to define a subset of the four signature gene categories to be used for its CGC predictions. The default value is only TC with CAZyme, and in my case I was primarily using CAZyme, TC, and TF as described in their paper. In both cases, when more than 2 STPs were encountered in between user-defined signature classes, CGCs were broken giving the illusion that dbCAN is predicting multiple CGCs, which were actually broken at random points.

In parallel, I also identified a number of bugs in PANORAMA systems detection module that were as detrimental to its predictions. Most notably, local filtering, a crucial step in the approach meant to only keep gene links that tend to be conserved across genomes, was never executed in the code. Moreover, the gene-level graphs used for projection of predicted systems to individual genomes were left without edges due to another bug, thus resulting in all projected systems appearing as split. Modification of certain lists while being used in a loop with multiple threading were also resulting in different outputs over runs, which were only stabilized with the definition of a hash seed for python.

In the lack of a clear description of PANORAMA systems detection approach, having ran into all these bugs, and struggling to make sense of the initial predicted systems, I decided to trace back the source code to reconstruct the approach, identify any additional bugs, and properly understand the generated output to be able to suggest ways to improve it. This allowed me to achieve a much clearer understanding of PANORAMA systems detection methodology and to identify potential areas for performance optimization or approach refinement, as further addressed in the Discussion section.

2.4 Systems Detection & Projection Methodologies

PANORAMA approach starts by constructing a pengenome-level context graph given the set of model gene families, i.e. all families included in the model and corresponding to HMM profiles used for annotation. To this end, linear contig graphs are first created to connect families within each contig. Subsequently, the combinations of all families found to be connected, i.e. within the user-defined transitivity distance of each other, are obtained. The context graph is defined as the aggregation of all contig graphs, but it is this set of combinations that will guide the prediction process, since each connected component of the context graph will be later intersected with each of these combinations to check for model rules verification. Notably, PANORAMA uses a local filtering approach based on a Jaccard-like index to only keep frequently occuring gene family connections across all strains that satisfy each given combination:

$$J_{\max}(g_1, g_2) = \frac{e_{g_1,g_2}(O)}{\max\left(f_{g_1}(O), f_{g_2}(O)\right)}$$

where:

$$
\begin{aligned}
O(g_1, g_2) &= \{\text{organisms } o \mid g_1 \in o \text{ and } g_2 \in o\} \\
e_{g_1,g_2}(O) &= \text{number of co-occurrences of } g_1 \text{ and } g_2 \text{ in } O \\
f_{g_1}(O) &= \text{number of occurrences of } g_1 \text{ in } O \\
f_{g_2}(O) &= \text{number of occurrences of } g_2 \text{ in } O
\end{aligned}
$$

Thus, depending on this filtering, either the whole combination will be validated as a system at the pangenome-level if it verifies the model rules, i.e. has the required families, or a subset of it will if it was fragmented by breaking an edge inside it. The complete pseudocode of the approach is provided in Algorithm 1.

After obtaining these species-level predictions, they must be projected for each of the strain genomes to separately detect their PULs. This is achieved by looping over the organisms in which a sufficient amount of a given system's families are found to verify the model rules, then constructing a gene-level projection graph for each of them. Subsequently, if the model families are found in close proximity forming an operon-like structure, i.e. if they meet the transitivity parameter constraint, the projected system is classified as strict; if they found on separate connected componenets within the gene-level graph, the system is classified as split; otherwise, the system is said to be extended. Therefore, the final output of PANORAMA includes one species-level systems.tsv file with all predicted PUL systems, then a seperate file for each strain for the projected systems verified in it. Notably, given its use of a window size of at least 1 (window size = transitivity +1), PANORAMA always detects non-model (also called non-signature) gene families in the vicinity of model families. In the final output, the two sets of model gene families and context (non-signature) families identified to be part of each system are presented separately.

2.5 PUL Resources & Comparative Analysis

The two publicly available PUL databases CAZy-PULDB [28] and dbCAN-PUL [21] were leveraged to validate PANORAMA PUL predictions and explore the potential of new PUL prediction rules formulation. In parallel, their corresponding genome-centric prediction tools, SusCD CAZy approach and dbCAN CGC-Finder, were used as benchmarks for evaluating the performance of our approach, thus addressing our second scientific question for this project Q2.

dbCAN CGC-finder was run with default parameters (distance of 2 and TC as signature gene category), then using both TC and TF as signature gene categories, on all genomes used to construct the B. thetaiotaomicron and P. denticola pangenomes, using their corresponding protein FASTA and GFF files downloaded from NCBI. As dbCAN output uses protein identifiers, it required proper mapping for comparison with PANORAMA output. This was achieved by matching the locus tag and protein id for all genes using the GFF files of all strains. Comparison of the predictions of both approaches with CAZy PULs was similarly achieved by properly mapping old locus tag identifiers as described above. The workflow of the latest dbCAN version is shown in Fig . 5. It begins with the prediction of CAZymes based on three parallel approaches that involve both sequence alignments with the CAZymes in CAZyDB and HMMs predefined based on the families in CAZyDB as well as their associated subfamilies. This is followed by CGCs prediction based on the previous CGC definition. dbCAN3 was further extended

with three approaches A, B, and C for substrate predictions based on reliable reported PULs, a BLAST against dbCAN-PUL database, and a majority voting approach, respectively.

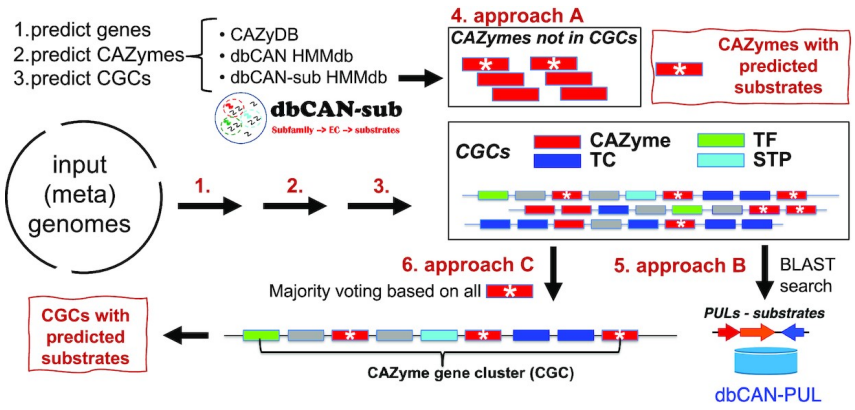


Figure 5: A diagram illustrating the workflow of dbCAN3. CAZymes prediction involves three parallel approaches based on sequence alignment with CAZyDB and pre-constructed HMMs for CAZyme families and subfamilies. This is followed with CGCs prediction following a relaxed set of rules introduced in dbCAN2. Finally, three parallel approaches are used for substrate predictions [26].

The lack of an API for CAZy-PULDB and its complex organization made automating its data accession quite challenging. This involved accessing each species predicted and empirically-validated PULs, then accessing each of their components genes to extract its locus tag. Since CAZy used old locus tag and did not provide genomic coordinates, the previously accessed GFF files of all genomes of each given species were used to match each locus tag to its new one and extract genomic coordinates. Finally, they were mapped back to their correspondig contigs and PUL ids. Notably, some of the old locus tags were not found in any of the GFF files. Similarly, while dbCAN provided a csv file of all their predicted PULs, amounting to a total of 673 empirically-validated PULs, this file only includes the CGC-Finder prediction of their components not the actually reported ones. So, dbCAN-PUL was website was similarly scrapped to extract PUL components. Overall, empirically-validated PUL data remain greatly lacking. Most of the data extracted from CAZy correspond to their predictions rather than actual PULs.

Given the scarcity of empirical data, comparative evaluations were conducted using literature-derived PULs when available, and consistently against CAZy and dbCAN CGC-Finder predictions to assess and validate PANORAMA's initial outputs. The examined comparison metrics included the overlaps between the sets of all genes predicted as PUL components, as well as the sensitivity of PUL systems detection examined by matching systems predicted in different methods and exanining the ratio of their overlaps to the size of the largest.

3 Results

3.1 Signature Families Annotation

In its current implementation, PANORAMA only supports gene families annotation using HMM profiles through pyHMMER. Thus, while it could annotate CAZymes, TFs, and STPs given their HMM profiles provided in dbCAN database, TCs annotations remain to be separately added. A total of 860 CAZyme, 240 TF, and 120 STP profiles were extracted from dbCAN HMM database. The family names used in these profile files were also extracted to be used in the model rules file or to be replaced by their corresponding class (CAZyme, TF, STP) in the annotation metadata written in the pangenome file. However, since dbCAN relies on three concurrent approaches for CAZymes annotation rather than just HMM profiles, I noticed significant discrepancies in the annotations performed by PANORAMA and dbCAN. An alternative is to annotate CAZyme gene families at the pangenome-level by running dbCAN on the FASTA file of all the protein families in the pangenome. These dbCAN annotations could then be added as metadata in the pangenome h5 file to be used for panorama systems detection similar to TC annotations obtained by running a DIAMOND search of the pangenome protein families against the pre-compiled TCDB dataset provided in dbCAN. Moreover, since PANORAMA does not support systems detection using multiple annotation sources, new annotations cannot be added to the ones previously done by PANORAMA pyHMMER. Instead, these annotations must be extracted from the pangenome binary file, combined with any new annotations, and formatted properly before being passed back to the pangenome file as annotation metadata. This is achieved by the metadata functionalities provided by PPanGGOLiN to allow extracting and writing metadata into pangenome binary files.

Considering these complications and to limit discrepancies at the annotation level and focus on comparing the predictions of PANORAMA and dbCAN given the same annotations, PANORAMA pyHMMER was discarded and dbCAN annotations of all signature families were added to the pangenome instead. However, dbCAN only annotates TFs, TCs, and STPs when run in its CGC-Finder mode, which requires gff files to extract coordinates, and thus could be only run for individual genomes. Therefore, all genomes used to construct the pangenome had to be accessed from NCBI to extract their corresponding protein FASTA and genomic GFF files. Subsequently, the gene-level annotations of dbCAN across genomes were merged together and mapped back to the family-level by matching each protein identifier to its corresponding gene family in the pangenome. A summary of these approaches is highlighted in Fig. 6.

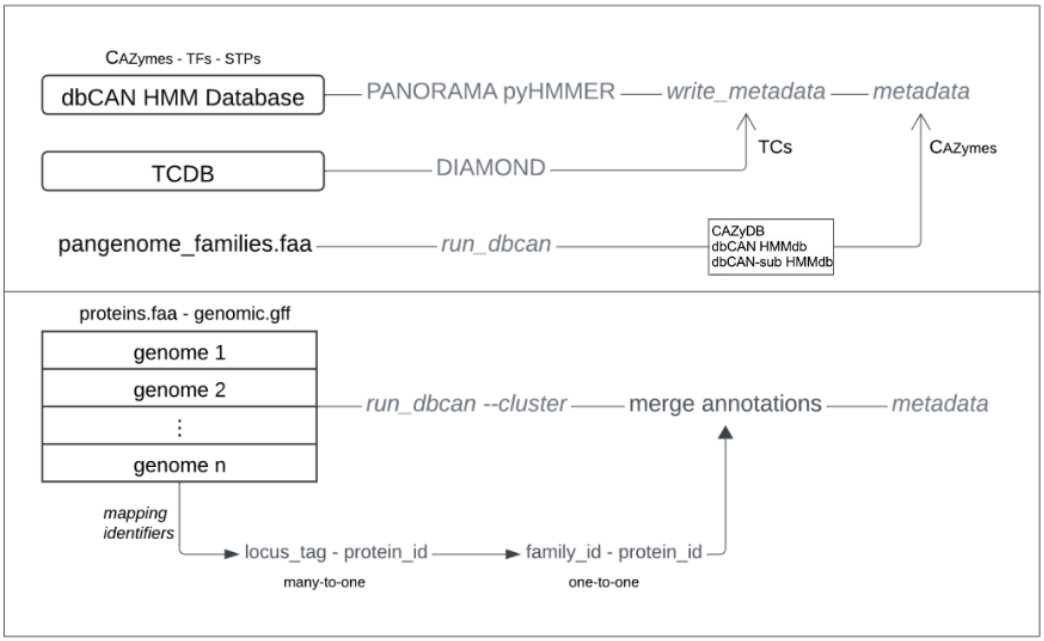


Figure 6: Representative diagram of the steps of the different annotation approach variations tested.

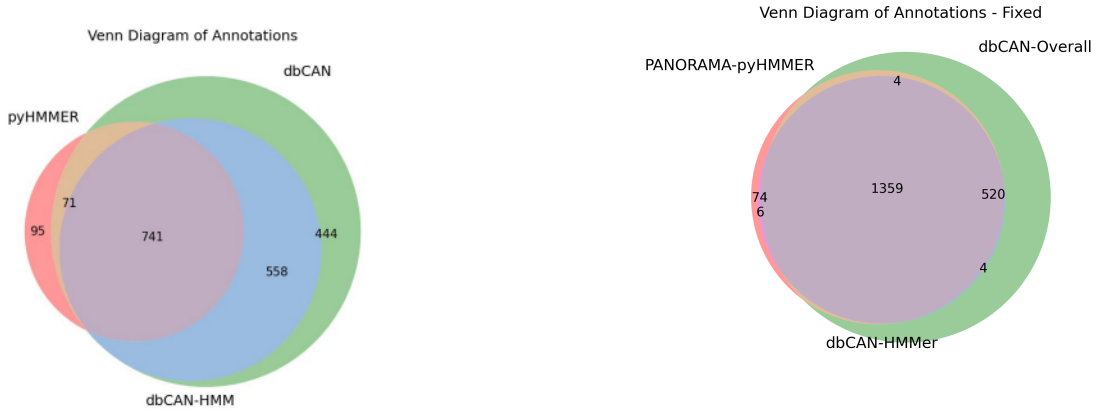


Figure 7: Comparison of CAZymes annotations across PANORAMA pyHMMER, dbCAN HMMer, and dbCAN combined annotations before (left) and after (right) fixing the annotation discrepancy issue.

Significant discrepancies were initially observed between the annotations obtained by merging dbCAN genome-level annotations and those obtained by PANORAMA PyHMMER and DIAMOND for TCs. The left-most venn diagram of Figure 7 highlights this discrepancy for CAZymes annotations between PANORAMA pyHMMER, dbCAN HMMer, and dbCAN overall predictions using its 3 concurrent approaches. Even after adjusting to the same parameters stated to be used in dbCAN. The initial hypothesis was that this could be due to the fact that dbCAN is examining all genes and thus finding much more matches that are missed when only examining the corresponding family as done in PANORAMA. This is because it is common for genes of the same family to correspond to different proteins. Therefore, mapping back protein annotations to families involves many-to-one matches. However, running PANORAMA annotation in sensitive mode, which runs PyHMMer across all genes of each family, took much

longer but only added very few more annotations. Further examination showed that in the vast majority of cases, all proteins of genes of the same family correspond to the same HMM profile across genomes and thus lead to one annotation at the family-level.

Instead, the observed discrepancy turned out to be due to a few parameters used in PANORAMA but not dbCAN, most notably the coverage parameter of 0.35 used in dbCAN corresponds to HMM profile coverage threshold but not target coverage which was set to 1. Similarly, a bias score threshold of 25 seems to be used in dbCAN since several annotations detected by PANORAMA at a score threshold of 0 but not in dbCAN were found to have bias score >25. Similarly, the DIAMOND evalue threshold used for TC annotation was found to be 1e-10, much higher then the 1e-102 cut-off used for CAZyme DIAMOND and stated on their webpage. This value was also identified by comparing DIAMOND results with those of dbCAN using a range of different cut-offs. The annotations overlap after these fixes is shown in the rightmost venn diagram of Fig. 7. Notably, several families were also found to match up to 3 different profiles of different classes. These include proteins acting as both STP and TF, TC and TF, or even STP, TF, and CAZyme. Moreover, I also noted more than 30 HMM profiles that were included in both the TF and STP HMM files extracted from dbCAN.

3.2 PUL Predictions & Validation

PANORAMA detection module was initially tested on the pangenome of *Prevotella denticola* constructed from 20 different strains. This species was selected for its relevance to the gut microbiome and its known PUL content, while also presenting a relatively simple pangenome with low PUL density—making it well-suited for initial testing. As no empirically-validated PULs were found for *P. denticola* or any other Prevotella species in either dbcan-PUL and CAZy-PULDB, PANORAMA predictions were constantly monitored in comparison with those of CAZy and dbCAN for individual genomes to guide the testing process and refine PANORAMA for improved performance.

Comparison with CAZy predictions was restricted to the only 7 *Prevotella denticola* strains found in CAZyDB. As highlighted in Fig. 8, PANORAMA tends to cover all genes across CAZy predicted systems and much more. The few genes identified by CAZy but not PANORAMA are mostly due to the issue of missing old locus tag identifiers discussed in 2.5. Notably, the opposite trend is observed when PANORAMA predicted context gene families are excluded and only model gene families are kept, as shown in Fig .9. This reveals that most of CAZy predicted genes do not fit within the signature gene categories defined in PANORAMA model, but are detected by PANORAMA since they tend to be in proximity of these signature genes. Moreover, the great decrease in the count of families predicted by PANORAMA between the two plots indicates that most of the families predicted as PUL components are non-signature families. Considering that these results are obtained at a transitivity of 0, it appears like PANORAMA's approach should be re-examined to somehow filter these context non-signature predictions.

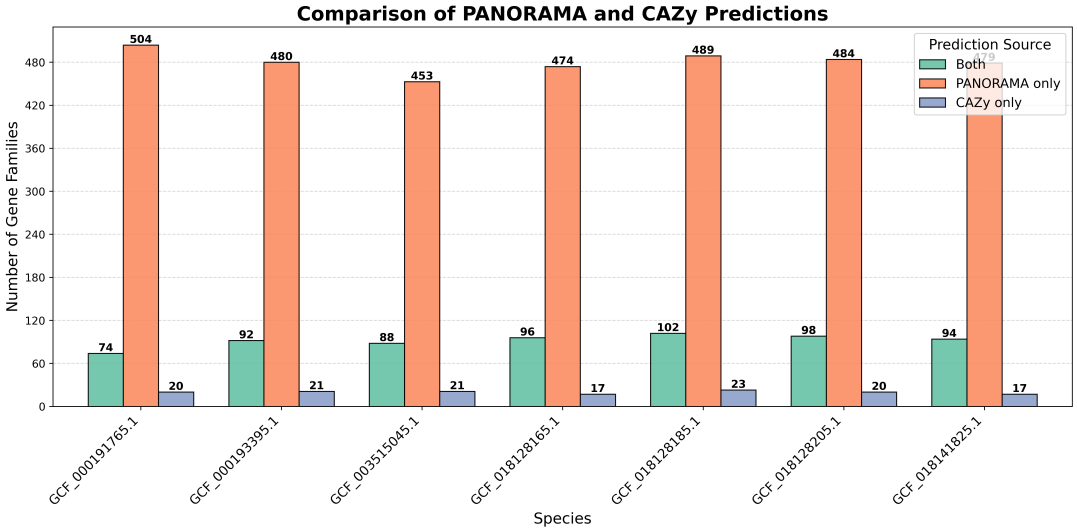


Figure 8: Frequencies of PUL gene components identified in both PANORAMA and CAZy and each separately.

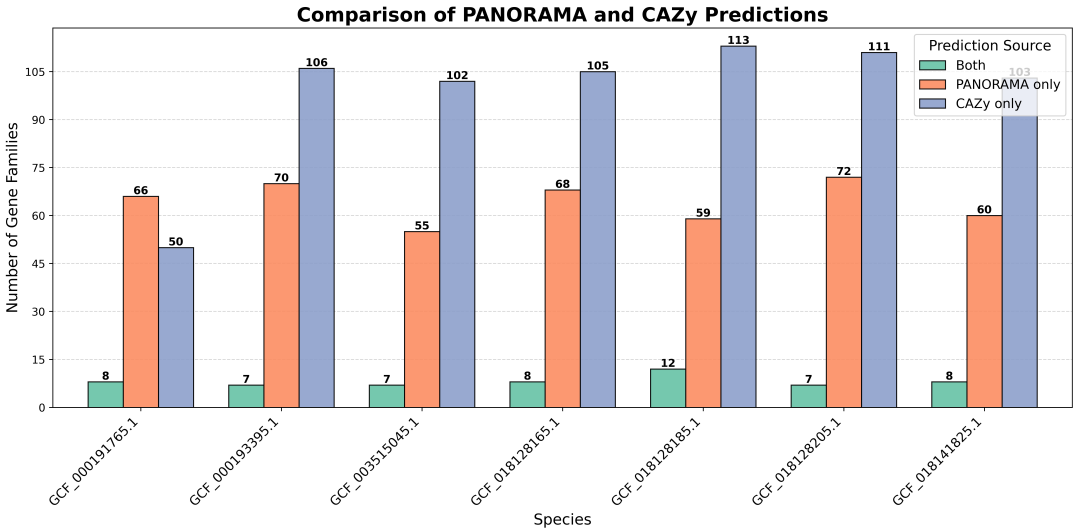


Figure 9: Frequencies of PUL model gene components identified in both PANORAMA and CAZy and each separately after filtering all non-signature genes.

Nevertheless, matching predicted systems between PANORAMA and CAZy versus dbCAN and CAZy reveals that dbCAN tends to predict even much longer CGCs as depicted in Fig. 10 across three genome instances of *P. denticola*. Nonetheless, the total number of genes across all PUL predictions tends to be much larger in PANORAMA, as shown in Fig. 11.

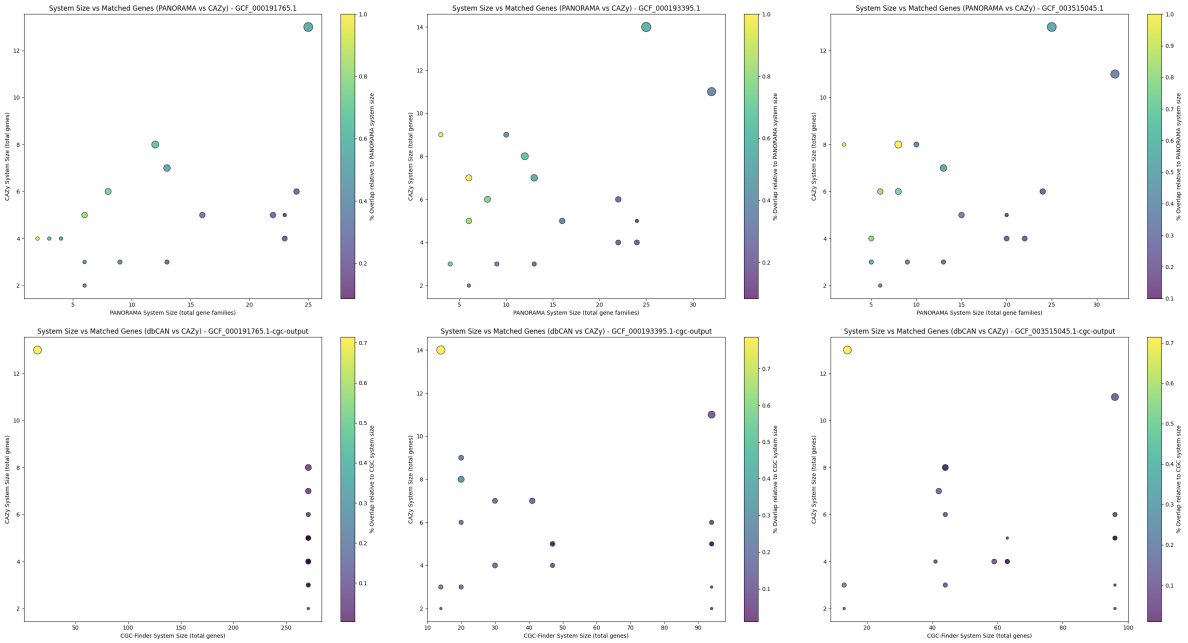


Figure 10: Overlap proportion of PANORAMA predicted systems and their corresponding ones in CAZy. Each dot represents one system matched between the two methods. The color is proportional to the extent of CAZy systems genes that are found in the corresponding PANORAMA prediction. The size is proportional to the number of genes per CAZy system. Note the scale differences between the plots for dbCAN (lower) and those for PANORAMA (upper

).

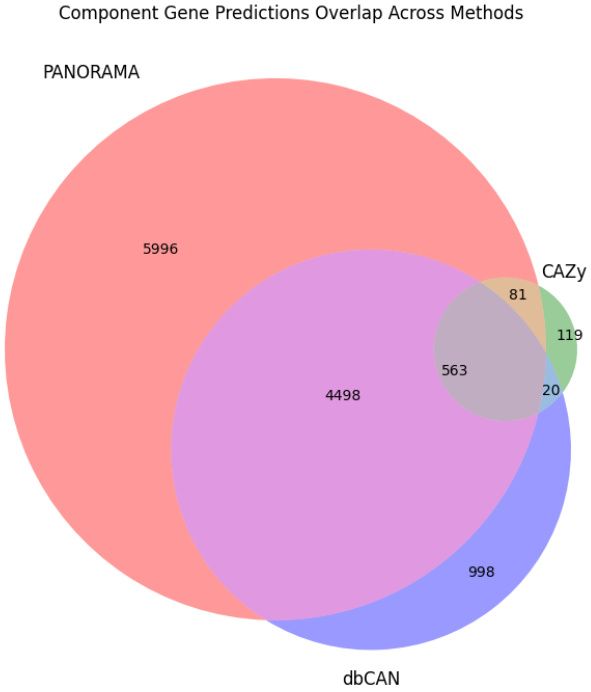


Figure 11: Venn diagram of the overlaps between the total counts of genes predicted as part of PULs by PANORAMA, dbCAN, and CAZy.

After achieving more promising predictions in *P. denticola*, PANORAMA was tested for

Bacteroidetes thetaiotaomicron pangenome constructed from a total of 327 strains to examine PANORAMA performance in a large pangenome highly rich in PULs and to benefit from the few empirically-validated PULs available to compare PANORAMA predictions to. Notably, empirical PULs in only two strains out of the 327 were identified. Their overlaps with PANORAMA predictions are highlighted in Fig. 12, showcasing a different trend compared to that noted for *P. denticola* in Fig 9. Notably, a significant proportion of the empirical PUL constituent genes is still detected by PANORAMA even after filtering out all context non-signature families. Moreover, the number of PUL constituent genes remained relatively high even after this filtering, which is likely related to the prevalence of PULs in Bacteroidetes making signature PUL families more abundant in them. While increasing transitivity to one improved sensitivity as shown in Fig. 23, it also led to a great increase in extra constituent genes by PANORAMA. It must also be noted that empirically-validated PULs tend to be larger than those predicted by CAZy Fig. 13, which could also play a role in the observed discrepancy.

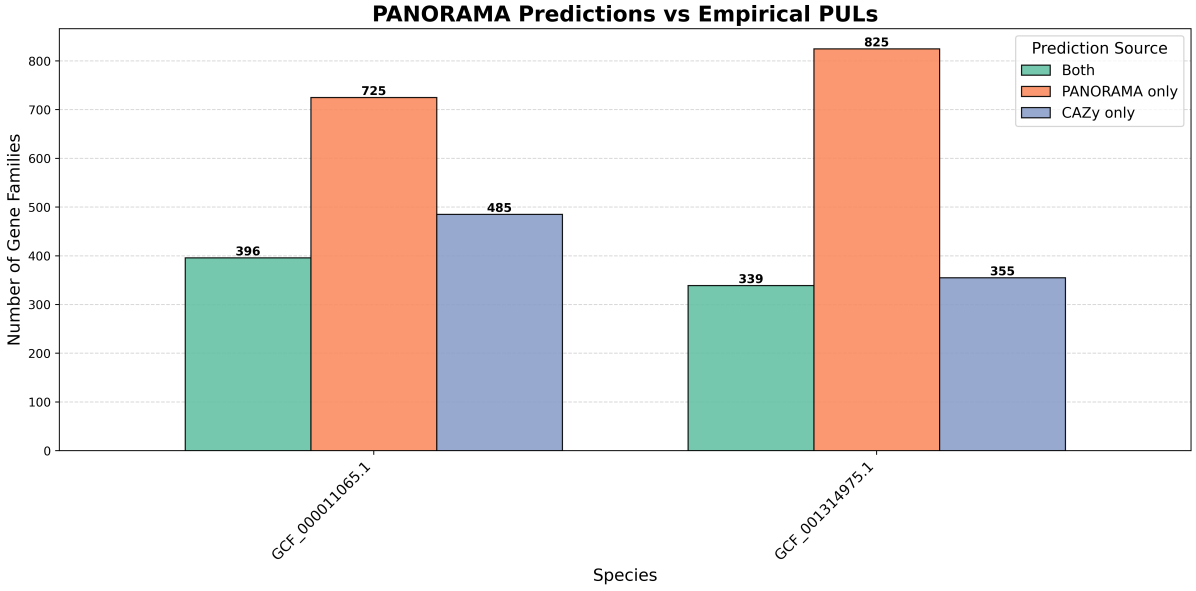


Figure 12: The frequencies of PUL model gene components identified by PANORAMA compared to empirically-validated PUL data extracted from CAZy-PULDB for two strains.

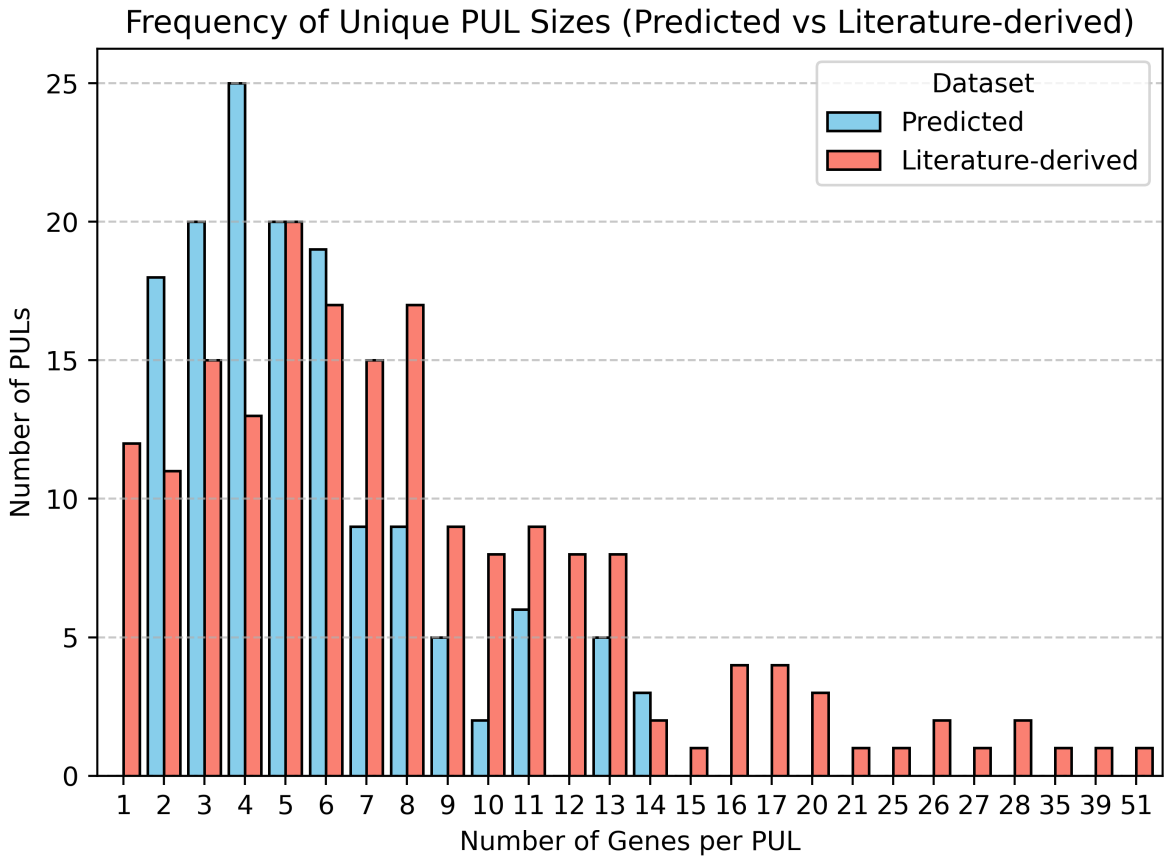


Figure 13: Frequency distribution of PUL sizes across strains for empirically-validated PULs versus CAZy predictions.

3.3 Performance Comparison

One important advantage for PANORAMA is its significantly enhanced running time. Figure 14 depicts the runtime performance of PANORAMA compared to dbCAN latest version across all pangenomes of Prevotella species. This performance comes at the cost of much larger memory costs spent on storing all the pangenomes data. Nonetheless, even the annotation step is much faster in PANORAMA since it is performed directly on gene families, whereas dbCAN must annotate all genes of each strain across species. Moreover, as highlighted in Fig. 15, the systems detection step is very fast detecting all species-level PULs for all 20 species in 3 minutes. The most time-consuming phase is that of projecting the predicted systems to individual genomes following their detection. However, this step could also be significantly optimized by avoiding the unnecessary use of gene graphs in the current implementation.

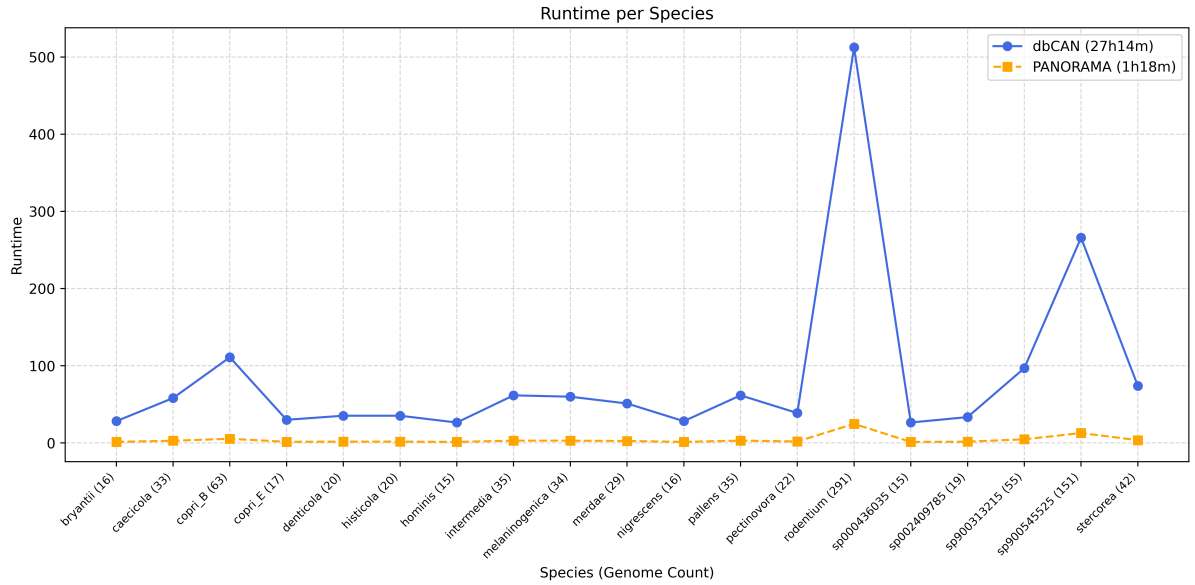


Figure 14: Running time fo PANORAMA vs dbCAN across all Prevotella species. The number of strains per species is indicated next to its name on the x-axis.

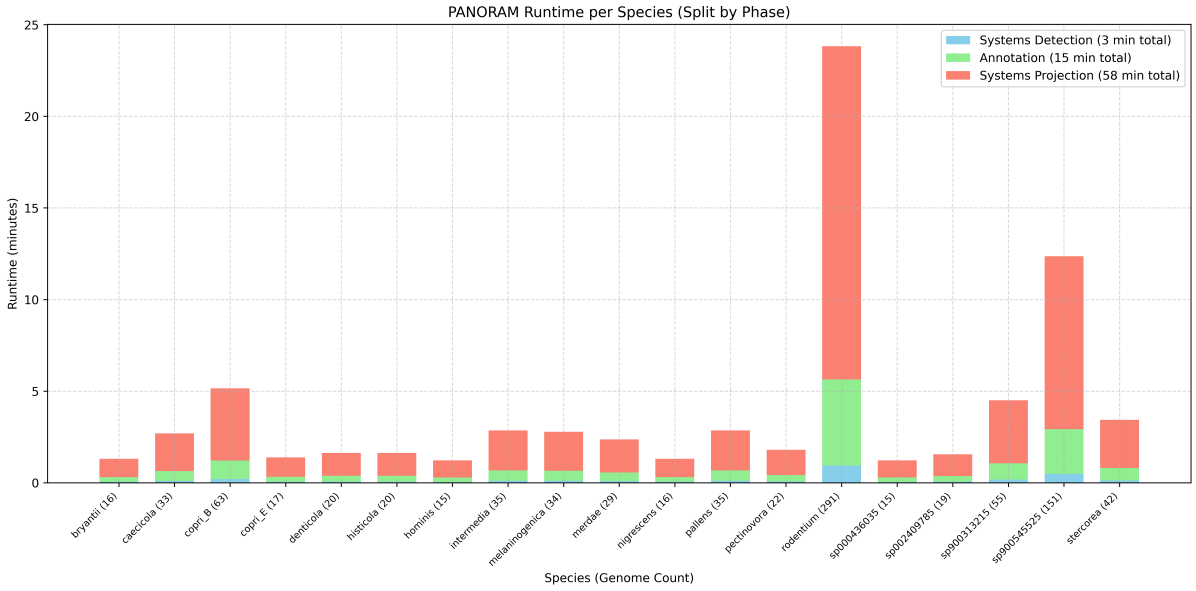


Figure 15: Running time of PANORAMA per species across its three phases.

Another important advantage of PANORAMA is its capacity to predict PULs regardless of genetic contiguity by benefitting from its pangenome-level predictions taking into account conserved gene family associations across all strains. Figure 16 highlights an example of a *P. denticola* strain with a fragmented contig-level assembled genome with more than 150 contigs. Expectedly, dbCAN failed to detected any CGCs in this genome, whereas as shown in the Proksee map, PANORAMA was able to detect up to 5 systems split across different contigs. Thus, PANORAMA was able to detect that certain pangenome-level PULs were present in this strain despite having their model families scattered in different contigs.

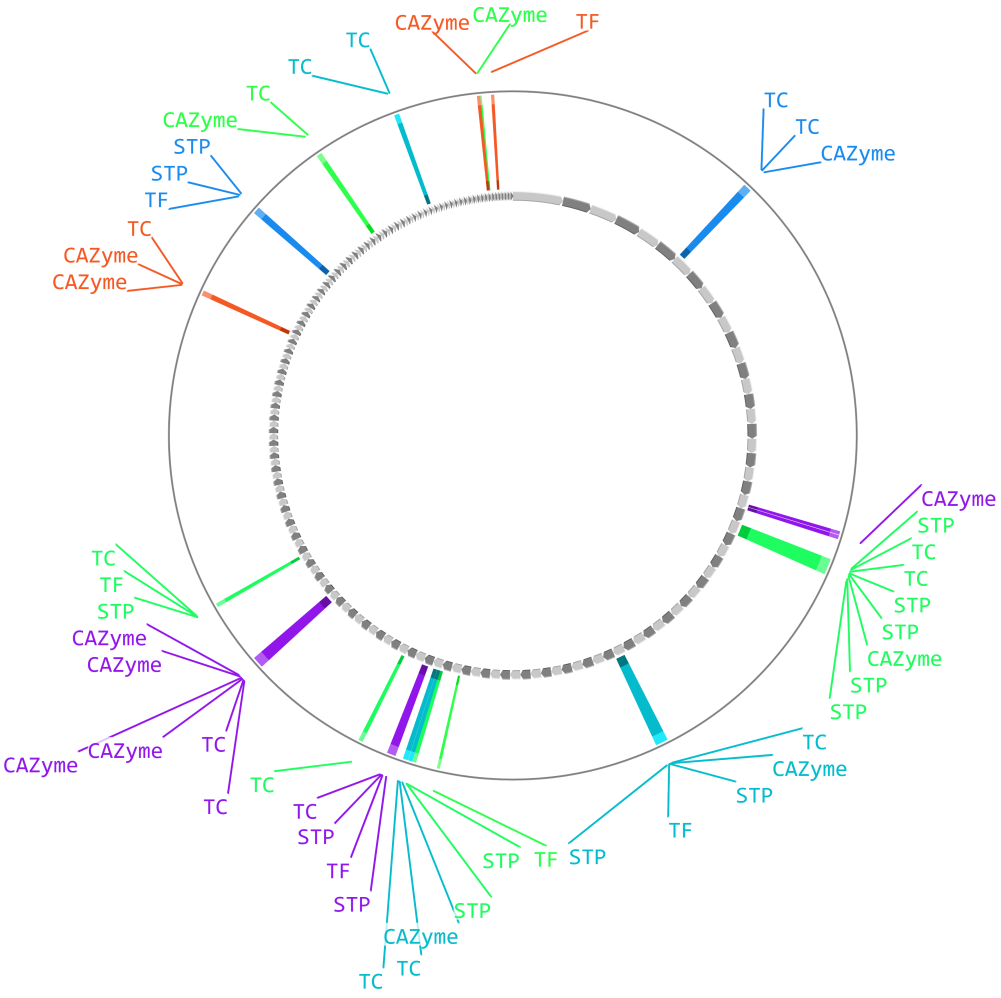


Figure 16: PANORAMA systems prediction for the fragmented genome case of the strain GCF_905372245.1 of *P. denticola* with more than 160 contigs in its assembly as could be seen in the central circle of the Proksee map. Five split systems could be observed, distinguished by different colors. The component model genes (CAZyme, TC, TF, STP) per PUL system are highlighted.

3.4 Model Refinement

While the adopted CGC-Finder prediction approach was modeled as a set of simple rules described in the methods section, it was clearly distinguishable from standard rule-based models as it was simply based on detecting one relevant feature then moving forward and backward from it to try to identify others within a user-specified distance. Typically, many rule-based models are defined to correspond to different variations of a given macromolecular system, with each of them specifying a set of few mandatory and accessory gene families instead of family classes that correspond to hundreds of families each. Therefore, I considered the possibility of defining multiple models based on different PUL characteristics and signature components. Yet, unlike other systems typically annotated with rule-based methods, such as defense systems, no previous reports on different PUL categories exist in the literature. Notably, MacSyFinder authors recognized that the applicability of their rule-based approach is dependent upon several

system-specific variables that determine the possibility of defining rules a priori, primarily the degree of conservation or evolution rate of the system. Thus, the high variability that seems to characterize PULs and the lack of empirically characterized PUL data play an important role in the unavailability of rules to properly identify and categorize them. The initial model developed by CAZy was based on the observation of the prevalence of the SusCD transport system pattern across experimental PULs from different Bacteroidetes species, and the dbCAN approach simply extended this definition to any transporter found near a CAZyme in an attempt to generalize the model beyond Bacteroidetes species.

As an initial attempt, I tried to use the 673 experimentally-validated PULs found in the latest version of dbCAN-PUL, last updated at the end of 2023, to identify potential patterns that could help create new PUL prediction rules. The conducted analysis can be found in the github repository: https://github.com/yazid-hoblos/PULkit. I started by grouping the PULs sharing the same substrate to identify the most common CAZymes per substrate group. Then, I identified single CAZymes and CAZyme pairs with a presence frequency exceeding a threshold (defined as half the size of a given group) as potential PUL signature elements. Using the identified signatures, simple models were created for the different substrate-based PUL groups, including up to 3 mandatory CAZymes and CAZyme pairs, as well as a minimum number of required CAZymes and a typical genomic size (taken as the median of all the PUL region sizes in a given group). In parallel, an RF model was also trained on the extracted PUL features to examine the extent of predictability in our data, and to compare its predictions to those achieved by the previously defined rules. This classifier simply uses the presence-absence patterns of all CAZyme families to predict the substrate of a given PUL. A comparison of the performance of the two approaches is presented in Fig. 17 and Table 18.

This initial attempt revealed the lack of clear consistent patterns to define multiple PUL models. Nonetheless, the Random Forest results highlight a great extent of predictibility in PUL empirical data, suggesting that with further refinements, well-distinguishable models could be created. Further exploration of this potential was left for later to focus on refining PANORAMA predictions given our initial CGC-Finder-like model first. More considerations regarding new models reformulation are addressed in the Discussion section.

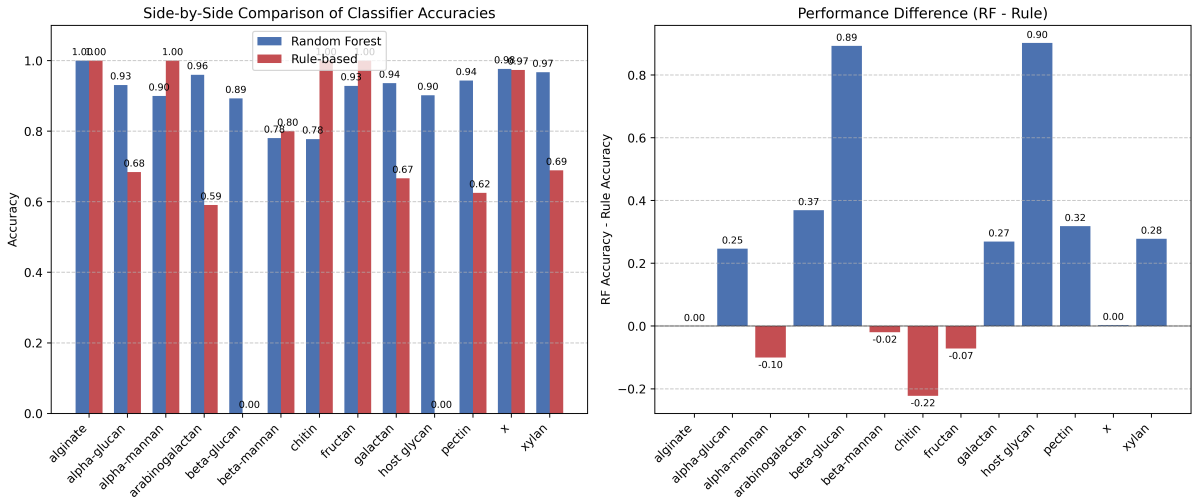


Figure 17: Comparative analysis of substrate prediction accuracy between an RF model and the formulated rules based on dbCAN-PUL empirical data.

Substrate	RF Accuracy	Rule Accuracy	RF (Correct/Total)	Rule (Correct/Total)	Better Method
alginate	1.000	1.000	23/23	19/19	Equal
alpha-glucan	0.931	0.684	54/58	26/38	Random Forest
alpha-mannan	0.900	1.000	9/10	10/10	Rule-based
arabinogalactan	0.960	0.591	24/25	13/22	Random Forest
beta-glucan	0.893	0.000	75/84	0/30	Random Forest
beta-mannan	0.780	0.800	32/41	20/25	Rule-based
chitin	0.778	1.000	14/18	5/5	Rule-based
fructan	0.929	1.000	13/14	11/11	Rule-based
galactan	0.936	0.667	44/47	12/18	Random Forest
host glycan	0.902	0.000	92/102	0/21	Random Forest
pectin	0.943	0.625	50/53	20/32	Random Forest
x	0.977	0.973	126/129	110/113	Random Forest
xylan	0.967	0.689	59/61	31/45	Random Forest

Figure 18: Associated comparative table highlighting the details of the predicted substrates by the two methods and the one with superior performance.

3.5 Postprocessing & Pan-Genus Insights

I noted that PANORAMA some pangenome-level predicted systems tend to be highly similar leading to an exaggerated number of PUL predictions, especially in bigger more complex pangenomes. For instance, more than 800 systems are predicted in the *B. thetaiotaomicron* pangenome of 327 strains. This is in part due to PANORAMA approach which takes into account all genome-level model families combinations with the potential of being predicted as a pangenome-level systems without any filtering. Therefore, if the same combination is found in multiple genomes with only one family difference between each, they will still be detected as separate systems leading to ambiguious results, especially when projected back to the individual genomes, since the same family will be found to be part of many pangenome-level systems. Indeed, for most *B. thetaiotaomicron* strains, several families are detected as part of up to more than 20 different systems. This prompted me to examine systems similarities at the pangenome-level. Figure 19 highlights the clustered heatmaps for both model and context gene families of of all 839 systems predicted for *B. thetaiotaomicron*. Many squares could be seen at the near the diagonal revealing high similarity between clusters of families as expected. Paradoxically, I also noted that in the case of *B. thetaiotaomicron*, context gene families were found to be very similar across all systems, which requires further examination. To further explore the potential of the clustering of these systems, a t-SNE dimension reduction was tested revealing up to 9 clusters as highlighted in Fig. 20.

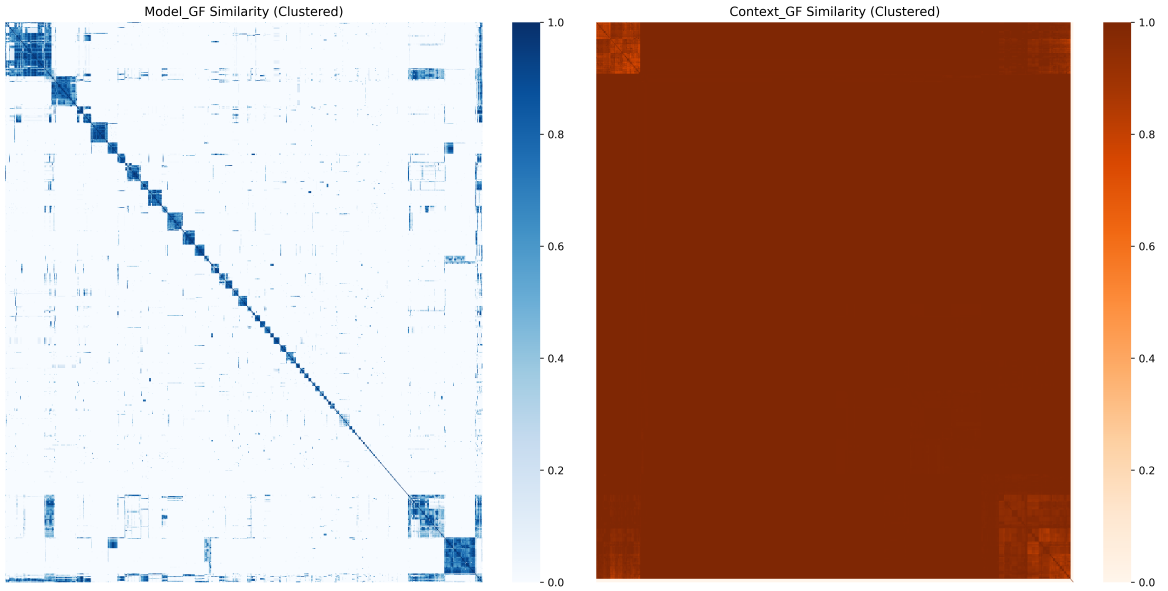


Figure 19: Clustered heatmaps of the model (left) and context (right) families of the 839 pangenome-level PUL systems detected by PANORAMA in *B. thetaiotaomicron*.

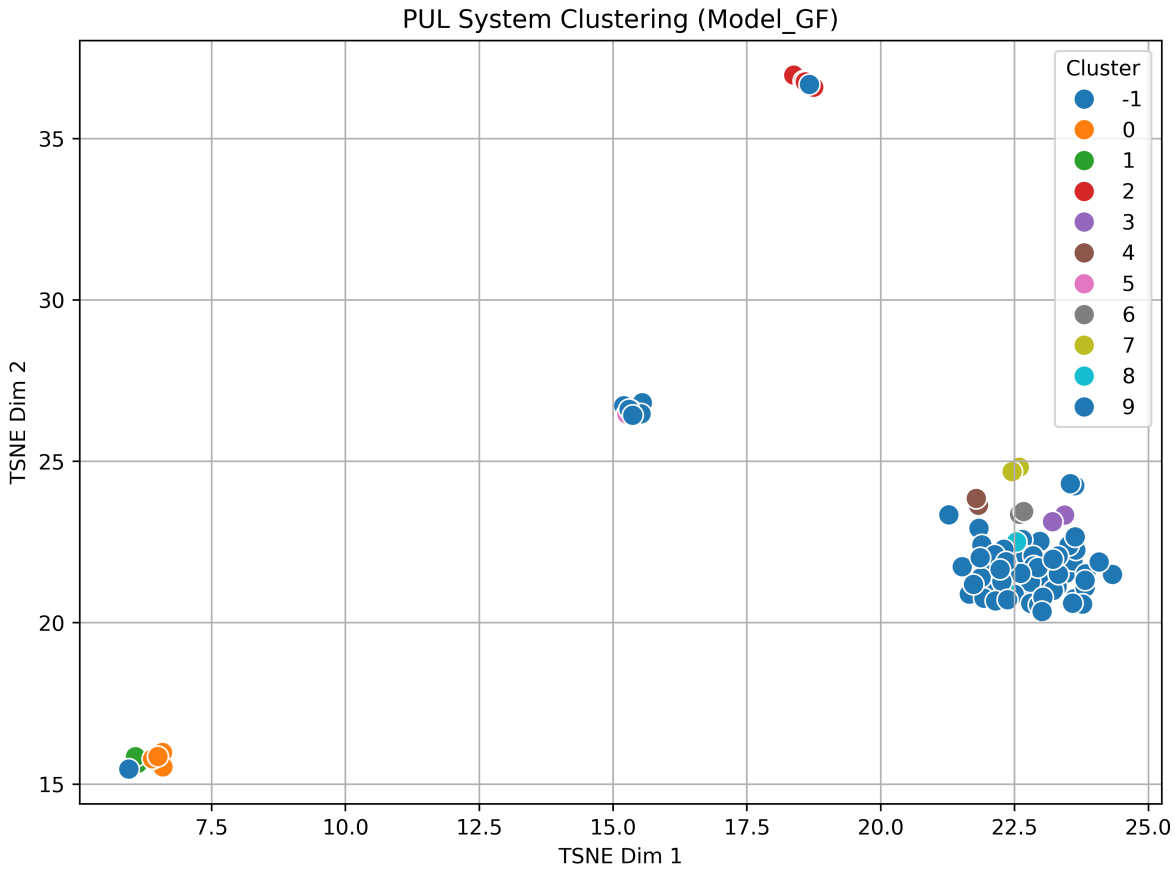


Figure 20: t-SNE dimensionality reduction results based on the model families of the systems detected by PANORAMA in *B. thetaiotaomicron*. A clustering algorithm was applied on the reduced dimension space revealing 10 different clusters. Cluster -1 refers to noise data not fitting any cluster (blue dots in the region centered around 22.5 on x-axis).

Furthermore, to gain insights regarding PUL systems conservation at the genus-level. All systems detected across all Prevotella species were examined for the potential of similar clustering. This first required the use of MMseqs2 to cluster gene families across all pangenomes. Subsequently, families within each system of each pangenome were mapped to their corresponding clusters and UMAP was conducted for all genus-level systems. As highlighted in Fig. 21, no clear seperation was noted across species systems, revealing that PUL architectures in Prevotella tend to be highly variable and lack conservation across species boundaries. However, as PANORAMA is still undergoing refinement for accurate PUL detection, this apparent lack of conservation may reflect limitations of the current model rather than a true biological absence. Notably, there were a number of small outliers detected in a region far away from all others (Figure 22), but further examination revealed that they simply corresponded to very small systems (3-5 families) across different species which appear to be found in one or very few organisms and thus appeared to be well-separated from all other systems.

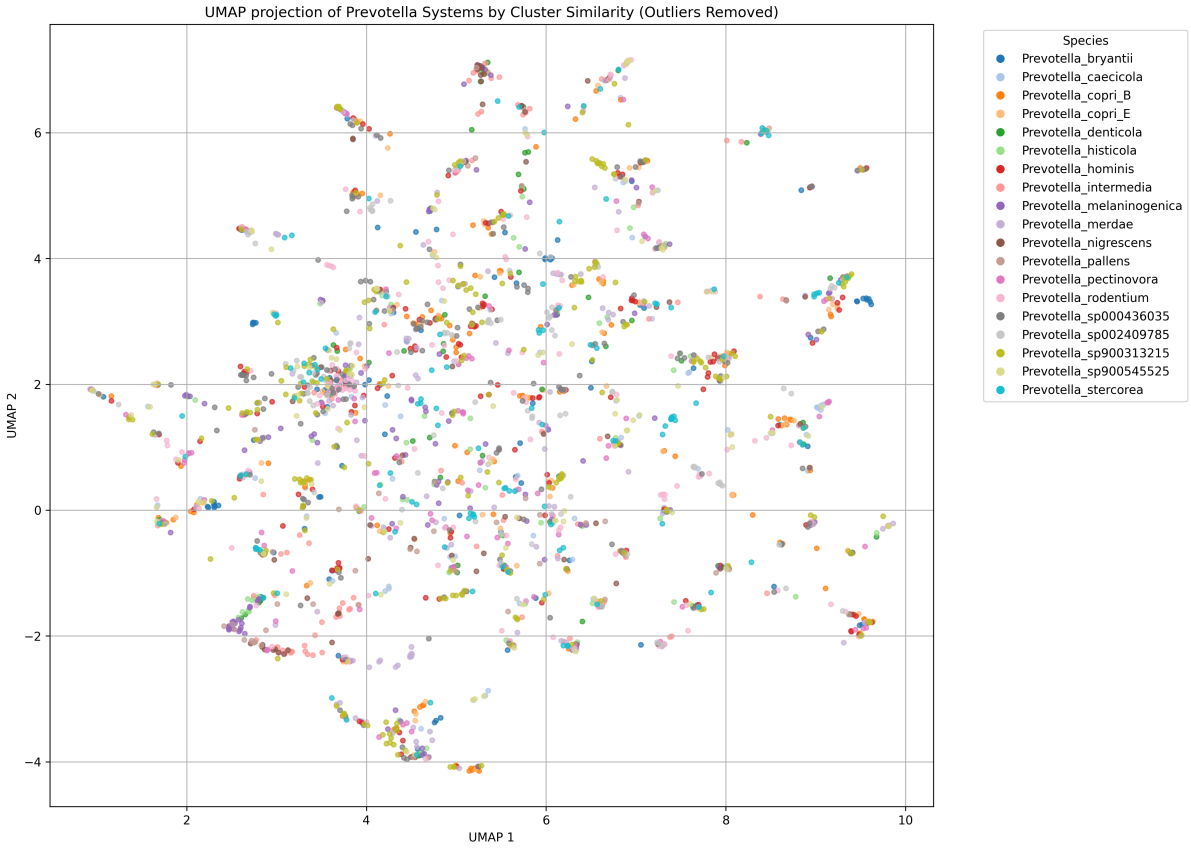


Figure 21: UMAP dimensionality reduction results across all 2319 systems detected across all species of Prevotella. Each dot represents a system colored by the species it was detected in. Families across systems were first clustered with MMseqs2 to allow for overlaps comparisons.

4 Discussion

The prediction of PUL systems presents a number of important challenges. Most notably, empirical data remain very scarce and current prediction approaches remain limited and relatively simplistic. Moreover, PULs display a great diversity both in sizes and content, exhibiting

different patterns across different species and genera,which makes the formulation of proper rules for their detection quite challenging. As PANORAMA was developed with defense systems in mind, it was tuned to their characteristic patterns, which differ markedly from those of PULs. Therefore, PANORAMA's predictions did not directly generalize to PULs and required adaptation and iterative refinement to capture their distinct structural and functional features. Nonetheless, PANORAMA initial predictions appear to be promising in several regards. Compared to dbCAN, PANORAMA demonstrates higher sensitivity, significantly faster runtime, and an improved ability to generalize to fragmented genomes. With further refinements, PANORAMA could pave the way for an unprecedented quality of PUL predictions. These refinements will be further explored in the remaining month of my internship, most notably at the level of models reformulations, approach optimization for PULs, and systems postprocessing.

The currently adopted model remains simplistic and somewhat arbitrary. Many experimentally-validated PULs do not seem to fit its requirements. In addition, more accessory signature genes could be defined, such as signal peptides, proteases, sulfatases, and polysaccharide-binding proteins (e.g. SusD). A separate model could be added for the traditional PULs fitting the Sus-CD-like pattern. Other models could also be considered based on PUL substrate classes, CAZyme subfamilies and functional units (linked to EC numbers). While thus o far, I have primarily only made use of the dbCAN-PUL CAZymes data to guide rules formulation, this could be further extended with their other signature components, notably TFs and TCs. To this end, I extracted these predicted features from the files of each PUL provided in dbCAN-PUL to create a dataset with PULs as rows and their predicted constituent elements as columns. For each element, its category (CAZyme, TF, TC, STP, or CDS) and annotation source are provided. Given the limited count of experimentally-reported PULs (only 673 in dbCAN-PUL), this analysis of PUL signatures could be also extended to predicted PULs. To this end, CazyDB and the recent dbCAN_seq database created from CAZymes and CGCs predictions in MAGs could be leveraged. However, relying on these predictions to identify new signatures risks being constrained to their own biases, especially considering the relative simplicity of their prediction rules. It could be also useful to extend the analysis to CAZyme subfamilies, as classified in dbCAN-sub based on eCAMI. Notably, dbCAN3 uses 3 parallel approaches for substrate prediction, one of them relies on subfamilies following the eCAMI approach: CAZyme subfamily $\rightarrow$ EC prediction $\rightarrow$ substrate. The second relies on BLAST against dbCAN-PUL, and the third is based on majority voting given the different predicted substrates found for the different CAZymes within a given CGC using the first subfamilies-based approach. Moreover, I will also try to benefit from consulting with field experts on previously reported PUL signatures and classes that could be modeled as MacSyFinder-like rules, and what approach to proceed with to define such models.

After going through PANORAMA source code, I noted that its current implementation could be significantly optimized, both in terms of runtime performance and approach enhancement. One important enhancement would be to discard gene-graphs used for systems projection as they simply correspond to linear connections and are never used except to check for connected components, which could be alternatively achieved through intervals definition. Furthermore, I noted that the context graph construction appears to be an unnecessary limiting step, since it is subsequently dismantled back to the intersections of each of its connected components with the combinations corresponding to contig-level sets of neighboring gene families. In fact, completely discarding the use of the context graph and directly using the small local aggregations of contig graphs for each combination led to the same predictions as the original

approach with faster performance. An interesting alternative would be to construct a systems graph that sequentially aggregates PULs as they are predicted, therefore reversing the approach from starting from a pangenome-level context graph and dismantling it to going from locally aggregated contig graphs to construct a pangenome-level systems graph encompassing all PULs for a given species. In addition, alternative filtering approaches could be explored, especially that I noted that in some instances, such as in the case of *P. denticola*, the same systems where detected before the bug resulting in local filtering never being executed was fixed. A final interesting refinement could be explored at the level of filtering the initial combinations used to guide the prediction process to avoid obtaining systems with high overlaps.

Alternatively, if approach-level refinements prove to be challenging and lead to unexpected behavior changes, post-processing PANORAMA predicted systems could be further explored as an alternative. As highlighted in the Results section, the potential for clustering similar systems was already established, allowing for better predictions interpretability and comparability. Finally, in the next steps, higher-level analyses could be also explored. For instance, after obtaining favorable results, the analysis could be extended to more species leveraging the *PanCompare* workflow of PANORAMA to get modules association to PULs and potentially proceed with MSA and phylogenetic reconstruction to try to trace back PULs evolutionary history.

5 Conclusion

In short, the potential of PANORAMA as an effective tool for predicting PULs was explored. Despite the limited availability of empirically-validated PUL data, PANORAMA showed promising performance by achieving higher sensitivity compared to existing tools like dbCAN, while also offering significantly faster runtimes and robust predictions in fragmented genomes. The approach of leveraging pangenome-level gene family associations enables PANORAMA to detect PULs even when genetic contiguity is disrupted. However, the current PUL prediction model remains simplistic and requires refinement to better capture the diversity and complexity of PUL architectures observed across species and genera. The high variability and lack of conserved signatures highlight the need for multiple, substrate-specific, or functional models rather than a one-size-fits-all approach. Preliminary machine learning attempts suggest that improved classification and model definition are achievable with further data and feature integration.

Future work will focus on refining prediction models by incorporating additional signature components and accessory genes, as well as leveraging expanded datasets including predicted PULs from metagenomic assemblies. Additionally, system postprocessing and clustering will be improved to reduce redundancy and enhance biological interpretability.

Overall, PANORAMA provides a strong foundation for advancing PUL prediction methodologies and offers a scalable framework adaptable to diverse microbial taxa, paving the way for deeper insights into carbohydrate metabolism in microbiomes. Over the course of my internship, my objective questions defined in the introduction were largely addressed: (1) I explored the reliability of PANORAMA PULs annotation at the pangenome-level; (2) I compared its predictions sensitivity and specificity to the two available genome-level approaches as well as empirically-validated PULs; (3) I tested PANORAMA in diverse species with varying characteristics and pangenome sizes ranging from a handful to hundreds of strains; and (4) I attempted to draw biological insights by analyzing predicted systems at the pan-genus level across all

Prevotella species.

References

[1] Hervé Tettelin, David Riley, Ciro Cattuto, and Duccio Medini. Comparative genomics: the bacterial pan-genome. *Current Opinion in Microbiology*, 11(5):472–477, 2008. ISSN 1369-5274. doi: 10.1016/j.mib.2008.09.006.

[2] Hervé Tettelin, Vega Masignani, Michael J Cieslewicz, Donati, et al. Genome analysis of multiple pathogenic isolates of *Streptococcus agalactiae*: Implications for the microbial 'pan-genome'. *Proceedings of the National Academy of Sciences*, 102(39):13950–13955, 2005. doi: 10.1073/pnas.0506758102.

[3] L Rouli, V Merhej, P-E Fournier, and D Raoult. The bacterial pangenome as a new tool for analysing pathogenic bacteria. *New Microbes and New Infections*, 7:72–85, 2015. doi: 10.1016/j.nmni.2015.06.005.

[4] Bo Segerman. The genetic integrity of bacterial species: the core genome and the accessory genome, two different stories. *Frontiers in Cellular and Infection Microbiology*, 2:116, 2012. doi: 10.3389/fcimb.2012.00116.

[5] Eugene V Koonin and Yuri I Wolf. Genomics of bacteria and archaea: the emerging dynamic view of the prokaryotic world. *Nucleic Acids Research*, 36(21):6688–6719, 2008. doi: 10.1093/nar/gkn668.

[6] Bruno Contreras-Moreira and Pablo Vinuesa. Get_homologues, a versatile software package for scalable and robust microbial pangenome analysis. *Applied and Environmental Microbiology*, 79(24):7696–7701, 2013. doi: 10.1128/AEM.02411-13.

[7] Gavin Tonkin-Hill, Nelle MacAlasdair, Christopher Ruis, Weimann, et al. Producing polished prokaryotic pangenomes with the panaroo pipeline. *Genome Biology*, 21:180, 2020. doi: 10.1186/s13059-020-02090-4.

[8] Andrew J. Page, Carla A. Cummins, Martin Hunt, Vanessa K. Wong, et al. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics*, 31(22):3691–3693, November 2015. doi: 10.1093/bioinformatics/btv421.

[9] Andries J. van Tonder, Surendra Mistry, James E. Bray, et al. Defining the estimated core genome of bacterial populations using a bayesian decision model. *PLoS Comput Biol*, 10 (8):e1003788, 2014. doi: 10.1371/journal.pcbi.1003788.

[10] Lars Snipen, Trygve Almøy, and David W. Ussery. Microbial comparative pan-genomics using binomial mixture models. *BMC Genomics*, 10:385, 2009. doi: 10.1186/1471-2164-10-385.

[11] Goran Rakocevic, Vladimir Semenyuk, Wen-Wei Lee, Spencer, et al. Fast and accurate genomic analyses using genome graphs. *Nature Genetics*, 51:354–362, 2019. doi: 10.1038/s41588-018-0316-4.

[12] Erik Garrison, Jouni Sirén, Adam Novak, Hickey, et al. Variation graph toolkit improves read mapping by representing genetic variation in the reference. *Nature Biotechnology*, 36:875–879, 2018. doi: 10.1038/nbt.4227.

[13] Guillaume Gautreau, Alexandre Bazin, Mathieu Gachet, Romain Planel, Laure Burlot, et al. Correction: Ppanggolin: Depicting microbial diversity via a partitioned pangenome graph. *PLOS Computational Biology*, 17(12):e1009687, 2021. doi: 10.1371/journal.pcbi.1009687.

[14] Paul Amours, Adelme Bazin, Guillaume Gautreau, Mathieu Dubois, Laura Burlot, et al. PanGBank: exploring GenBank genomes through a resource of pangenome graphs. In *JOBIM 2020 Journées Ouvertes Biologie, Informatique et Mathématiques*, Montpellier (virtual), France, June 2020. ⟨hal-04360000⟩.

[15] Adelme Bazin, Guillaume Gautreau, Claudine Médigue, David Vallenet, and Alexandra Calteau. panrgp: a pangenome-based method to predict genomic islands and explore their diversity. *Bioinformatics*, 36(Supplement_2):i651–i658, 2020. doi: 10.1093/bioinformatics/btaa792.

[16] Adelme Bazin, Claudine Médigue, David Vallenet, and Alexandra Calteau. panmodule: detecting conserved modules in the variable regions of a pangenome graph. *bioRxiv*, 2021. doi: 10.1101/2021.12.06.471380.

[17] Jose B. Pereira-Leal, Emmanuel D. Levy, and Sarah A. Teichmann. The origins and evolution of functional modules: lessons from protein complexes. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 361(1467):507–517, 2006. doi: 10.1098/rstb.2005.1807.

[18] Sophie S. Abby, Bertrand Néron, Hervé Ménager, Marie Touchon, and Eduardo P. C. Rocha. Macsyfinder: A program to mine genomes for molecular systems with an application to crispr-cas systems. *PLoS ONE*, 9(10):e110726, 2014. doi: 10.1371/journal.pone.0110726.

[19] Alon Zaslaver, Avi Mayo, Michal Ronen, and Uri Alon. Optimal gene partition into operons correlates with gene functional order. *Physical Biology*, 3(3):183–189, 2006. doi: 10.1088/1478-3975/3/3/003.

[20] Jie Feng, Yining Qian, Zhaoqing Zhou, Sydney Ertmer, et al. Polysaccharide utilization loci in bacteroides determine population fitness and community-level interactions. *Cell Host & Microbe*, 30(2):200–215.e12, 2022. doi: 10.1016/j.chom.2021.12.006.

[21] Catherine Ausland, Jinfang Zheng, Haidong Yi, Bowen Yang, Tang Li, Xuehuan Feng, Bo Zheng, and Yanbin Yin. dbCAN-PUL: a database of experimentally characterized CAZyme gene clusters and their substrates. *Nucleic Acids Research*, 49(D1):D523–D528, 2021. doi: 10.1093/nar/gkaa742.

[22] Jinfang Zheng, Le Huang, Haidong Yi, Yuchen Yan, Xinpeng Zhang, Jerry Akresi, and Yanbin Yin. Carbohydrate-active enzyme annotation in microbiomes using dbcan. *bioRxiv*, 2024. doi: 10.1101/2024.01.10.575125.

[23] Brandi L. Cantarel, Pedro M. Coutinho, Cédric Rancurel, Thierry Bernard, Vincent Lombard, and Bernard Henrissat. The carbohydrate-active enzymes database (cazy): an expert resource for glycogenomics. *Nucleic Acids Research*, 37(Database issue):D233–D238, January 2009. doi: 10.1093/nar/gkn663. Epub 2008 Oct 5.

[24] Yanbin Yin, Xizeng Mao, Jincai Yang, Xin Chen, Fenglou Mao, and Ying Xu. dbCAN: a web resource for automated carbohydrate-active enzyme annotation. *Nucleic Acids Research*, 40(W1):W445–W451, 2012. doi: 10.1093/nar/gks479.

[25] Han Zhang, Tanner Yohe, Le Huang, Sarah Entwistle, Peizhi Wu, Zhenglu Yang, Peter K. Busk, Ying Xu, and Yanbin Yin. dbcan2: a meta server for automated carbohydrate-active enzyme annotation. *Nucleic Acids Research*, 46(W1):W95–W101, 2018. doi: 10.1093/nar/gky418.

[26] Jinfang Zheng, Qiwei Ge, Yuchen Yan, Xinpeng Zhang, Le Huang, and Yanbin Yin. dbcan3: automated carbohydrate-active enzyme and substrate annotation. *Nucleic Acids Research*, 51(W1):W115–W121, 2023. doi: 10.1093/nar/gkad328.

[27] Nicolas Terrapon, Vincent Lombard, Harry J. Gilbert, and Bernard Henrissat. Automatic prediction of polysaccharide utilization loci in bacteroidetes species. *Bioinformatics*, 31 (5):647–655, March 2015. doi: 10.1093/bioinformatics/btu716.

[28] Nicolas Terrapon, Vincent Lombard, Elodie Drula, Pascal Lapébie, Saad Al-Masaudi, Harry J. Gilbert, and Bernard Henrissat. PULDB: the expanded database of Polysaccharide Utilization Loci. *Nucleic Acids Research*, 46(D1):D677–D683, 2018. doi: 10.1093/nar/gkx1022.

[29] Leighton J Payne, Sean Meaden, Mario R Mestre, Chris Palmer, Nicolás Toro, Peter C Fineran, and Simon A Jackson. Padloc: a web server for the identification of antiviral defence systems in microbial genomes. *Nucleic Acids Research*, 50(W1):W541–W550, 2022. doi: 10.1093/nar/gkac400.

[30] Florent Tesson, Antoine Hervé, Elodie Mordret, and et al. Systematic and quantitative view of the antiviral arsenal of prokaryotes. *Nature Communications*, 13:2561, 2022. doi: 10.1038/s41467-022-30269-9.

[31] Rob D. Stewart, Marc D. Auffret, Rainer Roehe, and Mick Watson. Open prediction of polysaccharide utilisation loci (pul) in 5414 public bacteroidetes genomes using pulpy. *bioRxiv*, 2018. doi: 10.1101/421024. Preprint.

[32] Fengfeng Mao, Parashuram Dam, Jun Chou, Victor Olman, and Ying Xu. DOOR: a database for prokaryotic operons. *Nucleic Acids Research*, 37(Database issue):D459–D463, 2009. doi: 10.1093/nar/gkn757.

[33] Benito Taboada, Karel Estrada, Rosa Ciria, and Enrique Merino. ProOpDB: Prokaryotic operon database. *Nucleic Acids Research*, 40(Database issue):D627–D631, 2012. doi: 10.1093/nar/gkr1020.

Appendix A: Approach Reconstructed Pseudocode

Algorithm 1: Context Graph Construction and Filtering

Input : Model families `model_fams`, window size `window_size`
Output : System units

```
foreach gene ∈ model_fams.genes do
    Add gene to contig_to_genes[gene.contig]
foreach (contig, genes) ∈ contig_to_genes do
    Initialize contig_graph as empty;
    foreach gene ∈ genes do
        foreach neighbor ∈ contig.genes[gene.position : gene.position + window_size + 1] do
            Add (gene.family, neighbor.family) to contig_graph
    Add contig_graph to context_graph;
    foreach cc ∈ contig_graph.connected_components do
        combs ← combs ∪ {cc ∩ model_fams}
foreach cc ∈ sorted(context_graph.connected_components) do
    cc_graph ← context_graph.subgraph(cc);
    cc_fams ← cc∩ model_fams;
    foreach comb ∈ combs do
        if comb ⊆ cc_fams then
            Remove cc_fams \ comb from cc_graph;
            local_filter(cc_graph);
            foreach sub_cc ∈ sorted(cc_graph.connected_components) do
                if check_rules(comb ∩ sub_cc) then
                    Add sub_cc to units
```

Appendix B: Supplementary Data

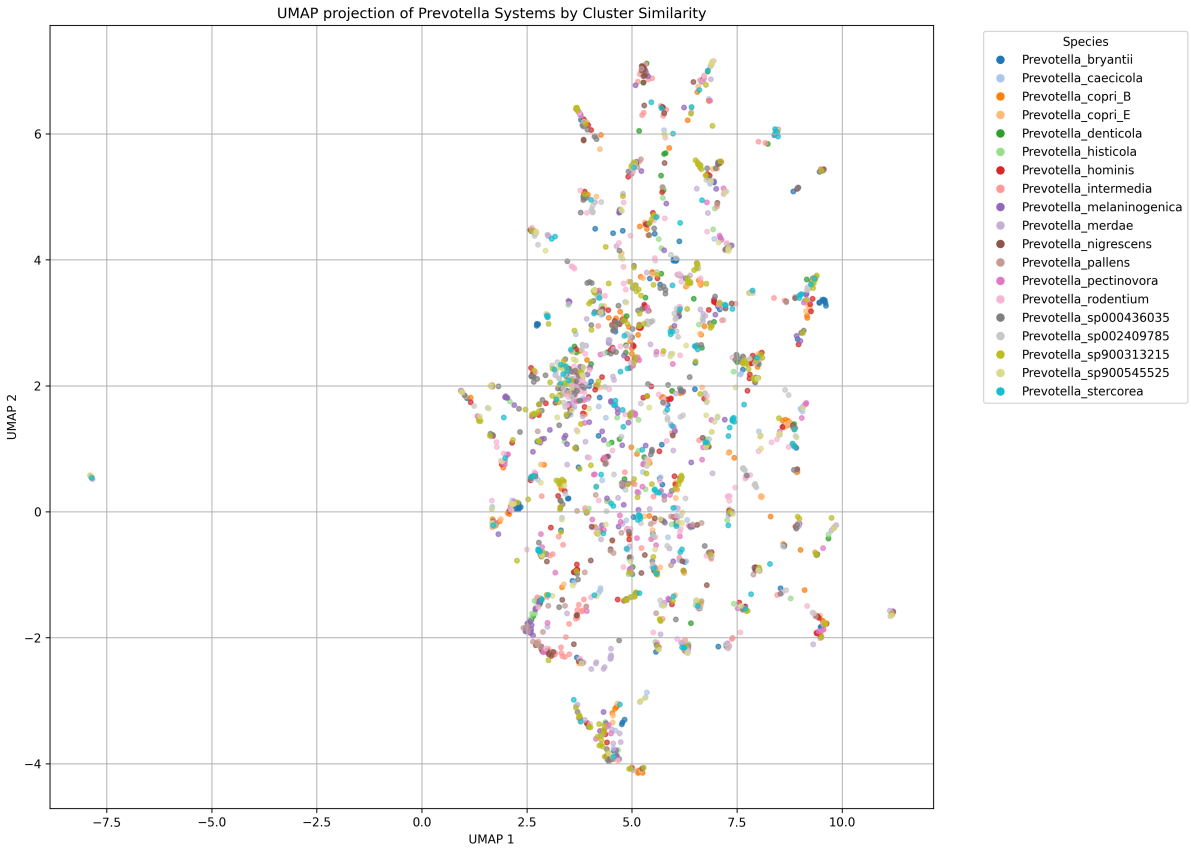


Figure 22: Unfiltered UMAP results across all systems of Prevotella.

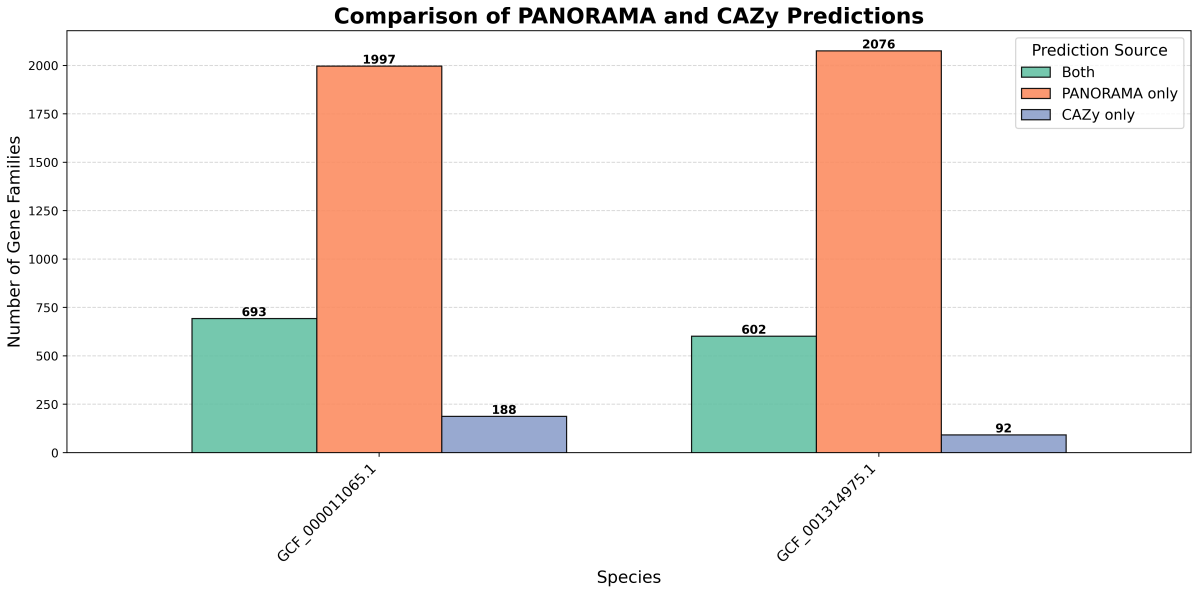


Figure 23: The frequencies of PUL gene components identified by PANORAMA compared to empirically-validated PUL data extracted from CAZy-PULDB for two strains. A transitivity of 1 was used and all families (model and context) were kept.

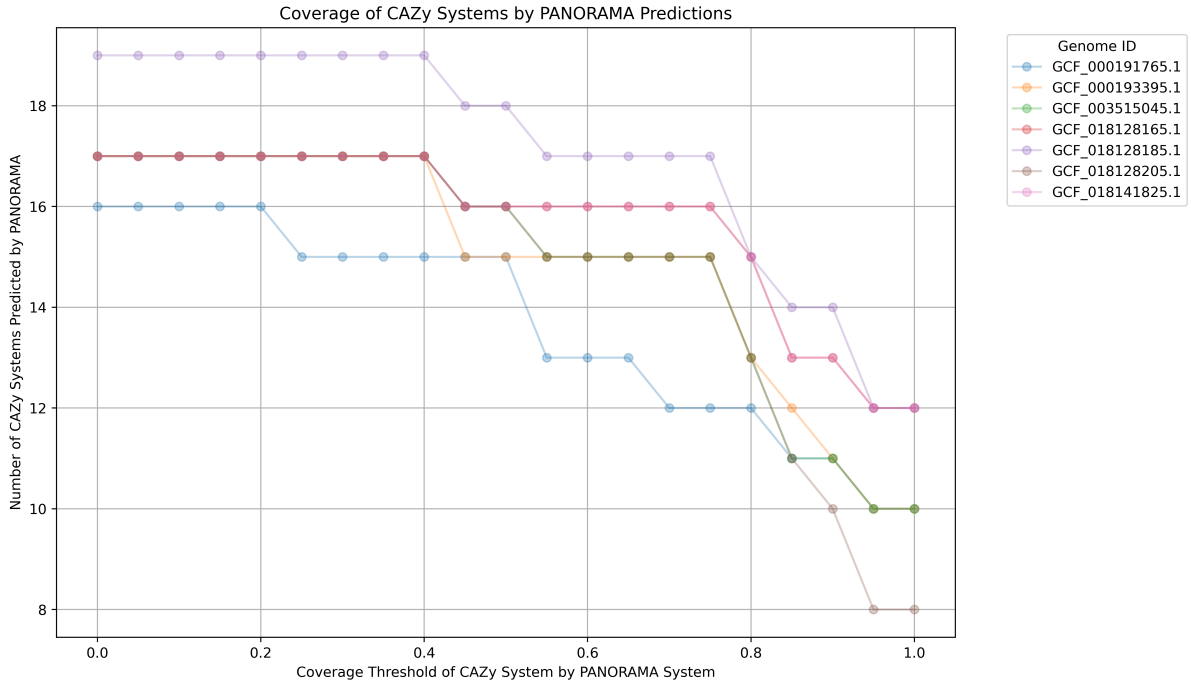


Figure 24: The number of CAZy systems covered by PANORAMA at different overlap thresholds for 7 strains of *P. denticola*.

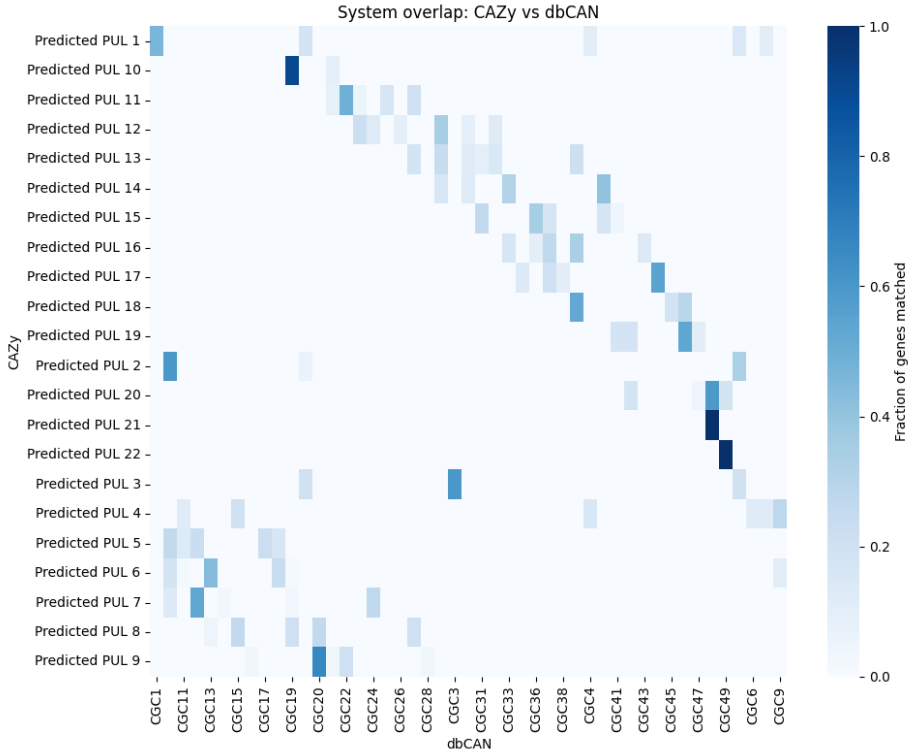


Figure 25: Heatmap of dbCAN vs CAZy systems predictions overlaps.

Table 1: Species and Strain IDs

Species	Strain IDs
Prevotella bryantii	GCF_000179055.1, GCF_022024255.1, GCF_905235005.1, GCF_900107655.1, GCF_022024215.1, GCF_947164775.1, GCF_946183655.1, GCF_002251425.1, GCF_900110745.1, GCF_946188295.1, GCF_022024235.1, GCF_019973295.1, GCF_900103315.1, GCF_002395335.1, GCF_000424605.1, GCF_022024195.1
Prevotella caecicola	GCF_948777195.1, GCF_947089325.1, GCF_949032535.1, GCF_949120255.1, GCF_948782355.1, GCF_949065795.1, GCF_948743705.1, GCF_948599345.1, GCF_947362355.1, GCF_949523975.1, GCF_949082815.1, GCF_947567555.1, GCF_948628575.1, GCF_948760135.1, GCF_947427255.1, GCF_910579315.1, GCF_948830945.1, GCF_949082625.1, GCF_948691275.1, GCF_948966255.1, GCF_948902085.1, GCF_949119285.1, GCF_949487315.1, GCF_948466625.1, GCF_949034885.1, GCF_949739765.1, GCF_949450365.1, GCF_947073065.1, GCF_947388975.1, GCF_949118995.1, GCF_948897545.1, GCF_013166515.1, GCF_009775545.1
Prevotella copri B	GCF_009494475.1, GCF_009495265.1, GCF_009493905.1, GCF_009495085.1, GCF_009494795.1, GCF_009494445.1, GCF_009493795.1, GCF_009494535.1, GCF_009495135.1, GCF_009494235.1, GCF_009493825.1, GCF_026015575.1, GCF_009494665.1, GCF_026015645.1, GCF_009494135.1, GCF_009495005.1, GCF_009494935.1, GCF_009494435.1, GCF_019249795.1, GCF_009494715.1, GCF_009495055.1, GCF_009494305.1, GCF_009495205.1, GCF_009494335.1, GCF_009493835.1, GCF_009495285.1, GCF_026015445.1, GCF_009494605.1, GCF_009495365.1, GCF_026015545.1, GCF_003471795.1, GCF_009494675.1, GCF_009494175.1, GCF_019249715.1, GCF_009495385.1, GCF_009494635.1, GCF_009495035.1, GCF_009493805.1, GCF_009493955.1, GCF_009494975.1, GCF_009494565.1, GCF_009494955.1, GCF_009494985.1, GCF_009493815.1, GCF_009493705.1, GCF_018784085.1, GCF_009494745.1, GCF_009494555.1, GCF_009494285.1, GCF_009495315.1, GCF_009495255.1, GCF_009495095.1, GCF_009493945.1, GCF_009494805.1, GCF_009494195.1, GCF_009494275.1, GCF_026015325.1, GCF_009494885.1, GCF_026015735.1, GCF_026015765.1, GCF_009493745.1, GCF_009494245.1, GCF_009495235.1
Prevotella copri E	GCF_003459205.1, GCF_024330415.1, GCF_026015775.1, GCF_024329915.1, GCF_024330565.1, GCF_024330625.1, GCF_024330465.1, GCF_024330785.1, GCF_024330385.1, GCF_026015565.1, GCF_003471065.1, GCF_024330605.1, GCF_003471085.1, GCF_003471195.1, GCF_024330865.1, GCF_026015465.1, GCF_024330575.1
Prevotella denticola	GCF_018141825.1, GCF_027682715.1, GCF_019391855.1, GCF_902483045.1, GCF_019375675.1, GCF_000421205.1, GCF_905373395.1, GCF_905372245.1, GCF_003515045.1, GCF_018128165.1, GCF_937990705.1, GCF_018128185.1, GCF_018128205.1, GCF_000193395.1, GCF_000759205.1, GCF_019375865.1, GCF_000191765.1, GCF_015262675.1, GCF_900454835.1, GCF_019375725.1
Prevotella histicola	GCF_018128125.1, GCF_015262395.1, GCF_019391885.1, GCF_019375695.1, GCF_015262615.1, GCF_000759075.1, GCF_019375895.1, GCF_015262355.1, GCF_019375835.1, GCF_019375655.1, GCF_015262445.1, GCF_015262435.1, GCF_019375975.1, GCF_019375585.1, GCF_018372295.1, GCF_015262545.1, GCF_916720365.1, GCF_015262625.1, GCF_000234055.1, GCF_019391935.1

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Table 1 – *Continued from previous page*

Species	**Strain IDs**
Prevotella hominis	GCF_959599415.1, GCF_004535825.1, GCF_900540745.1, GCF_019418425.1, GCF_003461825.1, GCF_022744025.1, GCF_934216725.1, GCF_022723075.1, GCF_022721035.1, GCF_022737195.1, GCF_022721995.1, GCF_022738855.1, GCF_025460925.1, GCF_021637945.1, GCF_959606135.1
Prevotella intermedia	GCF_002794335.1, GCF_002762425.1, GCF_902497285.1, GCF_002763975.1, GCF_000519165.1, GCF_002797175.1, GCF_003838285.1, GCF_002529225.1, GCF_002763745.1, GCF_002355195.1, GCF_001953935.1, GCF_002763845.1, GCF_001594285.1, GCF_023148185.1, GCF_002204405.1, GCF_002529435.1, GCF_002797155.1, GCF_000955645.1, GCF_003861325.1, GCF_002763785.1, GCF_002763715.1, GCF_000261025.1, GCF_002763575.1, GCF_002763955.1, GCF_002529305.1, GCF_002763535.1, GCF_001953955.1, GCF_001548195.1, GCF_000439065.1, GCF_002753835.1, GCF_002797135.1, GCF_003265025.1, GCF_900455005.1, GCF_002797185.1, GCF_002753855.1
Prevotella melanino-genica	GCF_000517825.1, GCF_019391735.1, GCF_015260455.1, GCF_002208725.2, GCF_946223055.1, GCF_015260375.1, GCF_018127905.1, GCF_938034465.1, GCF_015260465.1, GCF_951230405.1, GCF_019391775.1, GCF_015260555.1, GCF_015260265.1, GCF_019375815.1, GCF_000759305.1, GCF_015260405.1, GCF_019375485.1, GCF_015260495.1, GCF_015260395.1, GCF_015260355.1, GCF_019375935.1, GCF_015260635.1, GCF_901875335.1, GCF_019375525.1, GCF_020735905.1, GCF_019391815.1, GCF_015260605.1, GCF_018127885.1, GCF_905371445.1, GCF_019375605.1, GCF_020735785.1, GCF_000144405.1, GCF_905369805.1, GCF_902484395.1
Prevotella merdae	GCF_902475805.1, GCF_022717975.1, GCF_022719255.1, GCF_022731615.1, GCF_958450535.1, GCF_022722975.1, GCF_022721455.1, GCF_021622745.1, GCF_900542775.1, GCF_022721935.1, GCF_959597155.1, GCF_022730485.1, GCF_022747425.1, GCF_022733255.1, GCF_937983555.1, GCF_934196595.1, GCF_900290275.1, GCF_022738815.1, GCF_959598625.1, GCF_021614955.1, GCF_022737115.1, GCF_021621105.1, GCF_905209635.1, GCF_022721555.1, GCF_958445225.1, GCF_021628295.1, GCF_021622545.1, GCF_021631735.1, GCF_958349045.1
Prevotella nigrescens	GCF_902497265.1, GCF_018127845.1, GCF_031191185.1, GCF_938028575.1, GCF_905372985.1, GCF_000220235.1, GCF_915063465.1, GCF_018127865.1, GCF_002204475.1, GCF_018127825.1, GCF_000507825.1, GCF_019375595.1, GCF_015262115.1, GCF_900454985.1, GCF_000336235.1, GCF_019930605.1
Prevotella pallens	GCF_015261625.1, GCF_015261915.1, GCF_015261375.1, GCF_015261465.1, GCF_905371345.1, GCF_015261835.1, GCF_000220255.1, GCF_015261795.1, GCF_015261675.1, GCF_015261905.1, GCF_015261555.1, GCF_947062025.1, GCF_015261455.1, GCF_015261775.1, GCF_015261715.1, GCF_015261495.1, GCF_015261335.1, GCF_015261585.1, GCF_015261945.1, GCF_015262065.1, GCF_015261505.1, GCF_015261725.1, GCF_916720205.1, GCF_015261705.1, GCF_900454935.1, GCF_015261925.1, GCF_015261995.1, GCF_937889145.1, GCF_901875295.1, GCF_015261615.1, GCF_015261645.1, GCF_015261695.1, GCF_015261515.1, GCF_003269195.1, GCF_015262005.1

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Table 1 – *Continued from previous page*

Species	Strain IDs
Prevotella pectinovora	GCF_934141675.1, GCF_938034785.1, GCF_958438965.1, GCF_000833995.1, GCF_000833975.1, GCF_008680225.1, GCF_000834035.1, GCF_948470415.1, GCF_902469065.1, GCF_945788505.1, GCF_905204425.1, GCF_945905825.1, GCF_000833955.1, GCF_958441405.1, GCF_959603685.1, GCF_000436015.1, GCF_959026275.1, GCF_946410995.1, GCF_000834015.1, GCF_945947475.1, GCF_022770845.1, GCF_934167805.1
Prevotella rodentium	GCF_948765205.1, GCF_948752415.1, GCF_948547465.1, GCF_948478805.1, GCF_948734385.1, GCF_949011555.1, GCF_949542905.1, GCF_949106245.1, GCF_948692765.1, GCF_948867505.1, GCF_949429325.1, GCF_949082675.1, GCF_949071525.1, GCF_949003505.1, GCF_948755705.1, GCF_948606975.1, GCF_948848945.1, GCF_949100855.1
Prevotella sp000436035	GCF_934234285.1, GCF_022726665.1, GCF_022742045.1, GCF_022744985.1, GCF_958353465.1, GCF_902465045.1, GCF_945910485.1, GCF_938027345.1, GCF_000436035.1, GCF_959599205.1, GCF_905212065.1, GCF_934234225.1, GCF_958446975.1, GCF_934234255.1, GCF_900547875.1
Prevotella sp002409785	GCF_022643825.1, GCF_937910535.1, GCF_022654225.1, GCF_022644685.1, GCF_022642885.1, GCF_022642145.1, GCF_002397475.1, GCF_002409785.1, GCF_022648405.1, GCF_022483425.1, GCF_022646085.1, GCF_022653905.1, GCF_022646805.1, GCF_022486865.1, GCF_022487085.1, GCF_022653375.1, GCF_022482935.1, GCF_022649285.1, GCF_022483085.1
Prevotella sp900313215	GCF_958421975.1, GCF_021621995.1, GCF_022716495.1, GCF_021622165.1, GCF_959596635.1, GCF_022726855.1, GCF_958455005.1, GCF_958346525.1, GCF_009495355.1, GCF_022721195.1, GCF_003517095.1, GCF_022744555.1, GCF_947002785.1, GCF_022731225.1, GCF_022744255.1, GCF_021615575.1, GCF_958437225.1, GCF_958454265.1, GCF_019127485.1, GCF_022731475.1, GCF_022722775.1, GCF_959607805.1, GCF_900543835.1, GCF_022743305.1, GCF_900319145.1, GCF_019409885.1, GCF_019409665.1, GCF_022731675.1, GCF_958349985.1, GCF_959027125.1, GCF_019409165.1, GCF_902472465.1, GCF_022722175.1, GCF_958446805.1, GCF_958371865.1, GCF_022724235.1, GCF_022721515.1, GCF_029348455.1, GCF_905200085.1, GCF_000431915.1, GCF_937991355.1, GCF_019409625.1, GCF_021620165.1, GCF_019409485.1, GCF_022721595.1, GCF_021635845.1, GCF_022723795.1, GCF_934192035.1, GCF_958403555.1, GCF_022724485.1, GCF_022743545.1, GCF_934215665.1, GCF_022716095.1, GCF_022727795.1, GCF_022726655.1
Prevotella sp900545525	GCF_948978795.1, GCF_948564855.1, GCF_948972145.1, GCF_949533335.1, GCF_948937245.1, GCF_948722155.1, GCF_948702395.1, GCF_900545525.1, GCF_948775755.1, GCF_948506465.1, GCF_949527995.1, GCF_948778245.1, GCF_949015195.1, GCF_948629435.1, GCF_948486725.1, GCF_948656745.1, GCF_910583915.1, GCF_948531405.1, GCF_948880225.1, GCF_948892175.1

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Table 1 – *Continued from previous page*

Species	**Strain IDs**
Prevotella stercorea	GCF_958440665.1, GCF_959604165.1, GCF_959598925.1, GCF_958434195.1, GCF_958439675.1, GCF_022719095.1, GCF_021621265.1, GCF_000437535.1, GCF_959021395.1, GCF_022723675.1, GCF_958454795.1, GCF_021627625.1, GCF_958405245.1, GCF_022732015.1, GCF_021616235.1, GCF_022744465.1, GCF_022746025.1, GCF_905204625.1, GCF_021614855.1, GCF_000235885.1, GCF_016775315.1, GCF_022731075.1, GCF_022735295.1, GCF_958477515.1, GCF_958435615.1, GCF_938005225.1, GCF_022722935.1, GCF_021626785.1, GCF_021626105.1, GCF_022731575.1, GCF_022744865.1, GCF_021627665.1, GCF_022728525.1, GCF_022732275.1, GCF_021630825.1, GCF_958366955.1, GCF_023708285.1, GCF_022745005.1, GCF_958366335.1, GCF_021631635.1, GCF_022721255.1, GCF_022733235.1